Hirzebruch–Riemann–Roch Theorem

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August 23, 2025

Contents

In a sense, the introduction will pretend to be accessible, but we shall inevitably have to assume some background in a little bit of everything in geometry and algebra: Riemann geometry, complex and algebraic geometry, Basic K-theory and the theory of characteristic classes, and Elementary Hodge theory. The introduction can be thought of as me trying to make the ideas of the paper as accessible as possible to an undergraduate, but that goal has to drop as soon as we start proving things or else it will the paper will be hundreds of pages of background material.

Also, I chose to define the Chern character in a "classical" way as given by Hatcher [16] or Hirzebruch [7] instead of the "intersection theory" interpretation (for ex. given by [9]), however I will point out some interesting ways in which the two are linked (ex. a quick connection to Serge classes).

I also do a fundamentally different thing from Hirzebruch: besides being the literal opposite of "Bourbaki-dry" I also change the order to present the geometric material and χ_y-characteristic *first* as it totally motivates why the Todd class is defined in this context. Hirzebruch does the opposite because he has been infected with the Bourbaki-dryness that I want to avoid.

This paper will try to contain as many proofs as possible, but it is best considered as a foray of the path that build up to Hirzebruch Riemann Roch.

Ultimate Hirzebruch goal:

$$T_1 = \chi_1 = \tau$$

which is enough for them to agree.

Notation:

1. I would like that K or k represent a field, but K is already being used for $K(-)$ or the K-genus while k is often used for indexing and so it is too much, so let $\mathbb{F}$ represent an arbitrary field.

2. Let if $H_i^* \times H^{n-i}$ is a perfect pairing, as the manifolds/varieties we are working with are nice enough we shall abuse notation due to DeRham's Theorem and write:

$$\int_Y c$$

for the pairing of the cohomology class c with Y. If we want to make it more explicit, we shall write $\kappa_n[-]$ notation.

3. $A \sqcup B$ represents disjoint union

4. TM and T_M represent tangent bundles. $N_{M/N}$ represent the normal bundle of N with respect to M.

background:

1. Riemann geometry: Riemann metric, Levi-Civita connection, Gauss-Bonnet, signature, orientation. See [15]

2. cohomology theory: Hatcher level of knowledge ([4]), or see [6].

3. Theory of symmetric function (the basis that can be found in many group theory/field theory textbooks): see [14].

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Motivation: Riemann Roch Problem

A natural question to ask in complex and algebraic geometry is "how many holomorphic/meromorphic/algebraic/analytic/etc functions are there on a (suitable) space X". By letting $\mathcal{F}(X)$ represent our choice of functions (the global sections of a [quasi-coherent] sheaf $\mathcal{F}$), we shall denote this question as finding information about the 0th sheaf cohomological group

$$H^0(X, \mathcal{F})$$

which by [1] is equal to $\mathcal{F}(X)$. We denote $\mathcal{F}(X)$ in terms of sheaf cohomology because this group shall sometimes be trivial, and the reason for its trivialness can be described by higher order sheaf cohomology groups:

$$H^1(X, \mathcal{F}), H^2(X, \mathcal{F}), ...$$

exactly how to extract this information is known as the *Riemann Roch Problem.* To motivate the direction the problem took, let us answer this question in some simple cases, build up to how the Riemann Roch question is answer for algebraic curves and surfaces, and then motivate how the (classical) Riemann Roch Theorem would be generalized to a result about sections on vector bundles

1.1 Riemann Roch Problem

The simplest case would be to deal with polynomial functions over $X = \mathbb{C}$ which are determined by a finite set of roots (counting multiplicity) and a constant:

$$p(z) = c(z - a_1) \cdots (z - a_n)$$

and similarly for rational functions. For future connection to sheaves let $\mathcal{O}_X(X)$ represent all algebraic functions on X and $\mathcal{M}_X(X)$ represent all rational functions on X; other common notation is $\mathbb{C}[X] = \mathcal{O}_X(X)$ and $\mathbb{C}(X) = \mathcal{M}_X(X)$ where $\mathbb{C}$ is often replaced with some other field K or ring R. If we ask for holomorphic functions so that $(X, \mathcal{F}) = (\mathbb{C}, \mathcal{H}(\mathbb{C}))$, then, then the Weierstrass Factorization Theorem (see [17, section 4.2]) tells us that $f \in \mathcal{H}(\mathbb{C})$ factors as an infinite polynomial along with some correction factors

$$z^m e^{g(z)} \prod_k^{\infty} \left[\left(1 - \frac{z}{a_k}\right) e^{p_k(z)} \right]$$

where

$$p_k(z) = \frac{z}{a_k} + \frac{1}{2}\left(\frac{z}{a_k}\right)^2 + \cdots + \frac{1}{m_k}\left(\frac{z}{a_{m_k}}\right)^m$$

is the correction factor and the a_k are roots that form a non-accumulating sequence; in particular there are analytic concerns. A meromorphic function f on $\mathbb{C}$ can be determined by two entire function $f = g/h$ (again see [17, section 4.2]); this is possible as for poles of f we can define a function h with the appropriate roots and multiplicities. In other words, determining meromorphic functions becomes a classification of non-accumulating sequences in $\mathbb{C}$. The situation requires even more analytic information in hyperbolic space D^2 (think of the existence of $\sum_n z^{n!}$ and $1/(1-z)$). To understand the algebraic behaviour, limiting the scope to *compact spaces* such as Riemann surfaces or projective varieties gets much better as any holomorphic/meromorphic functions must have finitely many roots/poles by the Identity theorem (that if two functions agree on a set of accumulation points, they are equal), and so we can focus our efforts within the algebraic setting. This setting is plenty rich enough for study and often offers a good starting point to generalize to the non-compact case[1]. In fact, by the efforts of Serre, Chow, and Grothendeick, the category of analytic functions on (suitable) compact spaces is equivalent to the category of algebraic functions on these spaces ([18], [11]), meaning compact spaces are "intrinsically" algebraic. We shall prove some of the pertinent parts of this correspondence in ref:HERE.

Let us try finding holomorphic/meromorphic functions on $X = \mathbb{C}\mathrm{P}^1$ and see if they are determined by the points of $\mathbb{C}\mathrm{P}^1$ just like in the case of $\mathbb{C}$. Immediately, $\mathbb{C}\mathrm{P}^1$ has only constant holomorphic functions by an easy generalization of Liouville or the maximum modulus principle on manifolds. By the Residue Theorem, the sum of the number of roots and poles of a meromorphic function must be 0.

Let us represent this formally. Let

$$D = \sum_i n_i p_i \qquad p_i \in \mathbb{C}\mathrm{P}^1$$

be a finite formal sum; we may think of it as the free (abelian) group $F^{\mathbb{Z}}(\mathbb{C}\mathrm{P}^1)$. Call D a *divisor*, and let $\mathrm{Div}(\mathbb{C}\mathrm{P}^1)$ represent this group. A divisor can be thought of as an object that remembers which roots/poles with which multiplicity we are interested in. Define:

$$\mathrm{div}(f) = \sum_{p \in \mathbb{C}\mathrm{P}^1} \nu_p(f) p$$

where $\nu_p(f)$ is the order of the zero/pole, and is negative if it is a pole. This forms a subgroup of $\mathrm{Div}(\mathbb{C}\mathrm{P}^1)$; represent it as $\mathrm{PDiv}(\mathbb{C}\mathrm{P}^1)$ ("P" for principal), and elements of this subgroup are called

[1]Though these cases are still very much being studied, see ref:HERE

principal divisors. This subgroup represents the divisors that are *representable* as a meromorphic function. Let us find which divisors can be represented. Let:

$$\deg D = \sum_i n_i$$

Notice that $\deg(-)$ is certainly a group homomorphism. By our earlier remark:

$$\deg(\mathrm{div}(f)) = 0$$

which is a restriction applied to any meromorphic function on $\mathbb{C}\mathrm{P}^1$. It is easy to see that this restriction would work on any compact Riemann surface (see [17, chapter 6]). If $\deg(D) = 0$, by our earlier remark we have that there exists an f such that $\mathrm{div}(f) = D$, showing this conditions characterizes meromorphic functions $f \in \mathcal{M}(\mathbb{C}\mathrm{P}^1)$ (but it shall not on other Riemann surfaces). Now, consider the *(Weil) divisor class group*

$$\mathrm{Cl}(\mathbb{C}\mathrm{P}^1) = \frac{\mathrm{Div}(\mathbb{C}\mathrm{P}^1)}{\mathrm{PDiv}(\mathbb{C}\mathrm{P}^1)}$$

for any $[D] \in \mathrm{Cl}(\mathbb{C}\mathrm{P}^1)$, $\deg([D])$ is well-defined as $\deg(D) = \deg(D + f)$ for any $f \in \mathrm{PDiv}(\mathbb{C}\mathrm{P}^1)$. If $\deg(D) = \deg(D')$, then $\deg(D) - \deg(D) = \deg(D - D') = 0$, and hence $D - D' = \mathrm{div}(f)$. Therefore, we have that:

$$\mathrm{Cl}(\mathbb{C}\mathrm{P}^1) \cong \mathbb{Z}$$

which can be thought of as measuring the obstruction in a meromorphic function being defined by its points/roots. As sheaf cohomology measures obstruction of local information having global information, we shall have:

$$\mathrm{Cl}(\mathbb{C}\mathrm{P}^1) \cong H^1(\mathbb{C}\mathrm{P}^1, \mathcal{O}^*_{\mathbb{C}\mathrm{P}^1})$$

the details of this isomorphism and why $\mathcal{O}^*_{\mathbb{C}\mathrm{P}^1}$ shall be given in ref:HERE. The higher order sheaf cohomology groups shall be trivial as the dimension of $\mathbb{C}\mathrm{P}^1$ is 1 (over $\mathbb{C}$), and hence we'll have:

$$\begin{aligned} H^0(\mathbb{C}\mathrm{P}^1, \mathcal{O}_{\mathbb{C}\mathrm{P}^1}) &\cong \mathbb{C} \\ H^1(\mathbb{C}\mathrm{P}^1, \mathcal{O}^*_{\mathbb{C}\mathrm{P}^1}) &\cong \mathrm{Cl}(\mathbb{C}\mathrm{P}^1) \\ H^2(\mathbb{C}\mathrm{P}^1, \mathcal{O}(\mathbb{C}\mathrm{P}^1) &\cong \{0\} \\ &\vdots \end{aligned}$$

Translating the result for $(\mathbb{C}, \mathcal{M}_{\mathbb{C}})$, we see that:

$$\mathrm{Cl}(\mathbb{C}) = \frac{\mathrm{Div}(\mathbb{C})}{\mathrm{PDiv}(\mathbb{C})} \cong 0$$

which shall give us:

$$\begin{aligned} H^0(\mathbb{C}, \mathcal{O}_{\mathbb{C}}) &\cong \mathbb{C}[x] \\ H^1(\mathbb{C}, \mathcal{O}^*_{\mathbb{C}}) &\cong \{0\} \\ H^2(\mathbb{C}, \mathcal{O}(\mathbb{C}) &\cong \{0\} \\ &\vdots \end{aligned}$$

as every divisor can be represented by a principal divisor. We can interpret this as there being no impediments to using local information to construct meromorphic functions. Notice that the ring $\mathbb{C}[x]$ is a UFD; this algebraic fact shall be given geometric meaning in ref:HERE.

Let us look at the next simplest nontrivial example. Let $X = \mathbb{C}/\Lambda$ where $\Lambda \subseteq \mathbb{C}$ is a lattice spanned by the $\mathbb{R}$-linearly independent elements $(1, \tau)$ and let $\mathcal{F} = \mathcal{M}_X$ (all rational functions on X); X is called an *elliptic curve*, and so we shall label it as E. As E is compact, if $f \in \mathcal{M}_E$ then $\mathrm{div}(f) = 0$. However, if $\deg(D) = 0$, this does not imply there exists an $f \in \mathcal{M}_E$ such that $\mathrm{div}(f) = D$. To see this, let's analyze $\mathcal{M}_E$ a little more closely. As $\mathbb{C}/\Lambda$ is the quotient by Λ, any meromorphic function must respect the Λ structure:

$$f(z + n + m\tau) = f(z) \qquad \forall n + m\tau \in \Lambda$$

such a function is called *doubly periodic* or *elliptic function* (as they are defined on elliptic curves). Using this periodic condition, it is simple to see that $\mathrm{div}(f) = 0$ by looking at:

$$\frac{1}{2\pi i}\int_{\partial D} \frac{f'(z)}{f(z)} dz = 0$$

where D is the fundamental domain[2], namely the opposing edges cancel out. We can modify this integral to find other properties, for example:

$$\frac{1}{2\pi i}\int_{\partial D} z\frac{f'(z)}{f(z)} dz = n + m\tau \in \Lambda$$

which is nonzero as z is not doubly periodic. Replacing z by other algebraic functions all end up in Λ, and so we see that if $D = \mathrm{div}(f)$:

$$\sum_i n_i p_i \equiv 0 \bmod \Lambda$$

where here $\sum_i n_i p_i \notin F^{\mathbb{Z}}(E)$ but $\sum_i n_i p_i \in E$, namely we are treating p_i as a complex number multiplied by n_i and summed. Certainly there is no such restriction on a general divisor $D \in \mathrm{Div}(E)$ of degree 0. We thus have shown that there is a surjective map:

$$\mathrm{Cl}^0(E) = \frac{\mathrm{Div}^0(E)}{\mathrm{PDiv}(E)} \twoheadrightarrow \mathbb{C}/\Lambda \tag{1.1}$$

where $\mathrm{Div}^0(E)$ are all divisor of degree 0. It is in fact an isomorphism, which comes from constructing an elliptic functions using the Weierstrass $\wp$-function, the details are in [17, chapter 5]. Thus, we get the short exact sequence:

$$0 \to \mathbb{C}/\Lambda \to \mathrm{Div}(E) \xrightarrow{\deg} \mathbb{Z} \to 0$$

often called the *Jacobi-Abel* exact sequence as $\mathbb{C}/\Lambda$ is a Jacobian variety, and a short exact sequence:

$$0 \to \mathrm{Cl}^0(E) \to \mathrm{Cl}(E) \xrightarrow{\deg} \mathbb{Z} \to 0$$

All together, we get a short exact sequence and a commutative diagram:

$$0 \to \mathrm{PDiv}(E) \to \mathrm{Div}^0(E) \to E \to 0$$

[2] If there are zero/poles on the path, we simply shift around appropriately

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{Div}^0(E) \arrow[r, hook] \arrow[d, two heads, "\pi"] & \mathrm{Div}(E) \arrow[r, "\deg"] \arrow[d, two heads, "\pi"] & \mathbb{Z} \arrow[r] \arrow[d, equal] & 0 \\
0 \arrow[r] & \mathrm{Cl}^0(E) \arrow[r, hook] & \mathrm{Cl}(E) \arrow[r, "\deg"] & \mathbb{Z} \arrow[r] & 0
\end{tikzcd}$$

Continuing on to other Riemann surfaces, we shall loose the isomorphism given in equation (1.1) and is the beginning of the study of Jacobian varieties (see [17, section 6.4]). Going in higher dimensions is the beginning of the study of Weil and Cartier Divisors and more generally Chow Groups (see [1]), and study how far away a space is from being constructed by "principal" (codimension 1) objects, namely the study of $H^1(X, \mathcal{O}_X)$[3]. Overall, we get some useful information about he obstruction to finding global sections, but not necessarily concrete ways of finding $H^0(X, \mathcal{O}_X)$.

To answer the Riemann Roch problem, a more granular approach to find which functions (of various kinds) exist on a space X must be taken. As divisors keep track of root/pole information, they are a great tool in classifying functions that "satisfy" the properties of the divisor. To make this precise, define an order on divisors $\mathrm{div}(X)$ where (for now) X is either a Riemann surface or a smooth projective curve. Define a partial order on $\mathrm{Div}(X)$ by saying $D_2 \geq D_1$ if each coefficient of $D_2 - D_1$ is non-negative. Then we can consider:

$$\mathcal{O}_X(D)(X) = \{f \in \mathcal{M}_X(X)^* \mid \mathrm{div}(f) + D \geq 0\}$$

In other words, if $D = \sum_i n_i p_i$, then $\mathcal{O}_X(D)(X)$ is all meromorphic functions f where it has a pole at p_i of order at most n_i if $n_i > 0$ and roots at p_i of order at least $|n_j|$ if $n_j < 0$ (and $\mathcal{O}_X(D)$ is a [coherent] sheaf). From this, we can ask the Riemann Roch problem for $H^0(X, \mathcal{O}_X(D))$. The famous Riemann-Roch theorem as proven by Riemann and Roch (where Riemann proved the Riemann-inequality, and Roch provided the necessary work to make it an equality, see [13]) is now known as *Riemann Roch for curves* and is the equality:

$$\dim H^0(X, \mathcal{O}_X(D)) - \dim H^1(X, \mathcal{O}_X(D)) = \deg(D) + 1 - g \tag{1.2}$$

where g is the genus of the Riemann surface or smooth projective curve (see [19] or [1] for defining the genus of a curve via scheme theory, or for a direct computational proof using normalization see [17]). In the original phrasing of the Riemann Roch theorem, the group $H^1(X, \mathcal{O}_X(D))$ is replaced with $H^0(X, \mathcal{O}(K_X - D))^\vee$ where K_X is the canonical divisor of X; these groups are isomorphic by Serre Duality (see [1, section 6.3]). Then this space is identified with the space of meromorphic 1-forms (either through some simple techniques from complex analysis, see Brill-Noether Reciprocity in either [13] or [17], or by combining of Poincaré duality with DeRham's Theorem, see either [8] or [15]). Switching to the traditional notation $l(D) = \dim H^0(X, \mathcal{O}_X(D))$ and $i(D) = \dim H^0(X, \mathcal{O}_X(K_X - D))^*$ we get:

$$l(D) - l(K_X - D) = \deg(D) + 1 - g \qquad \text{(equiv.)} \qquad l(D) - i(D) = \deg(D) + 1 - g$$

This result is proven in theorem 7.1.1. As we phrased the Riemann-Roch problem in the beginning, this formula would then be asking for the value of $l(D)$. There are cases where $l(K - D)$ disappears (for example if $D = 0$ or $D = K$), and there are many cases we can compute it easily enough. It is simple to compute the right hand side but in general it is harder to computer the left hand side. Thus, this formula becomes a new constriction that helps us find $l(D)$. There are a set of theorems known as *vanishing theorems* that specify under which conditions $l(K-D)$ (or more generally, higher cohomology groups) vanish (see [12]); under these conditions we exact recover $l(D)$.

[3]Notice the object are codimension 1, so the "dual" have dimension 1

Let us work towards the intuition behind the generalization. The notation on left hand side in equation (1.2) can be compactified by using the Euler-Poincaré characteristic (or simply the Euler characteristic): $\chi(X, \mathcal{O}_X(D)) = \sum_{i=0}^{\infty}(-1)^i \dim H^i(X, \mathcal{O}_X(D))$ so that:

$$\chi(X, \mathcal{O}_X(D)) = \deg(D) + 1 - g$$

As X is a curve $H^i(X, \mathcal{O}_X(D)) = 0$ for $i > 1$, cite? Reference for later? giving the equality. The right hand side shall be drastically different, split into special invariants of vector bundles known as the Chern class and Todd class where the Chern class can be thought of as capturing some "algebraic" or "local" information (this will generalize the $\deg(D)$) while the todd class shall capture the "geometric" or "global" information (this will generalize the $1+g$). In fact, the Euler Characteristic can easily be generalized to be an invariant on vector bundles by using sheaf cohomology instead of singular cohomology, and hence the Riemann Roch theorem generalizes to a question about the existence of sections on vector bundles over a compact manifolds, namely if $(X, \mathcal{F})$ is a compact manifold along with a vector bundle $\mathcal{F}$, then we want to compute:

$$H^0(X, \mathcal{F})$$

The result of this will be:

$$\chi(X, \mathcal{F}) = \int_X \mathrm{ch}(X)\mathrm{td}(X)$$

This equation is precisely the result of the *Hirzebruch-Riemann Roch Theorem*. We shall mention the generalization to smooth quasi-projective schemes over a field near the end, and also talk a bit about a recent paper ([2]) generalizing to finite dimensional Noetherian schemes.

How do Vector Bundles enter the Picture

To motivate how vector bundles will enter the picture, it is worth-while seeing how $\mathcal{O}_X(D)$ can be interpreted as a line bundle. To that end, recall that the projective curve $X = \mathbb{P}^1$ as defined in classical algebraic geometry only has constant global sections. Namely, if $U_0, U_1 \cong \mathbb{A}^1$ is the usual cover with global sections (i.e. functions) $\mathcal{O}_X(U_0) \cong \mathcal{O}_X(U_1) = K[x]$ and the compatibility condition on $U_0 \cap U_1$ is $f(z) = g(1/z)$, namely:

$$\begin{tikzcd} U_0 & U_1 \\ U_0 \cap U_1 \arrow[u, hook] \arrow[r, "\cong"] & U_0 \cap U_1 \arrow[u, hook] \end{tikzcd} \quad \text{i.e.} \quad \begin{tikzcd} \mathbb{A}^1 & \mathbb{A}^1 \\ \mathbb{A}^1 \setminus \{0\} \arrow[u, hook] \arrow[r, "\cong"] & \mathbb{A}^1 \setminus \{0\} \arrow[u, hook] \end{tikzcd}$$

gives:

$$\begin{tikzcd} \mathcal{O}_X(U_0) \arrow[d] & \mathcal{O}_X(U_1) \arrow[d] \\ \mathcal{O}_X(U_0 \cap U_1) \arrow[r, "\cong"] & \mathcal{O}_X(U_0 \cap U_1) \end{tikzcd} \quad \text{i.e.} \quad \begin{tikzcd} K[x] \arrow[d] & K[y] \arrow[d] \\ K[x, 1/x] \arrow[r, "\cong"] & K[y, 1/y] \end{tikzcd}$$

where $x \mapsto 1/y$ and $1/x \mapsto y$, Then if $F([x' : y']) \in \mathcal{O}_{\mathbb{P}^1}(\mathbb{P}^1)$ is a global function where $x = x'/y'$ and $y = y'/x'$ (when $y', x' \neq 0$ respectively), then $F([x : 1]) \to f(x) \in \mathcal{O}_X(U_0)$ and $F([1 : y]) \to g(y) \in \mathcal{O}_X(U_1)$ are the restrictions of the function to U_0 and U_1 respectively, and by the above diagram it must be that these functions map to the same element in the restriction, namely:

$$f(x) \to f(x) = g(1/x) = g(y) \leftarrow g(y)$$

but it is impossible that $f(x) = g(1/x)$ in $k[x]$ unless f, g are the same constant. Thus, it must be that the only global functions on the projective line are constants:

$$\mathcal{O}_{\mathbb{P}^1}(\mathbb{P}^1) = K$$

The fundamental limitation here is that the restriction $f(x) = g(1/x)$ is very strong. What if we can loosen this restriction? For example what if

$$f(x) = xg\left(\frac{1}{x}\right)$$

Then we can in fact form linear homogeneous functions! For example consider

$$F([x' : y']) = ax' + by'$$

Then F maps to $f(x) = ax + b$ and $g(y) = a + by$. Then:

$$f(x) = ax + b = x\left(a + b\frac{1}{x}\right) = xg\left(\frac{1}{x}\right)$$

which works! More generally, if the condition $f(x) = x^n g(1/x)$ allows for degree n homogeneous polynomials, and the homogeneity can happen for any number of variables. Denote the degree n-homogeneous polynomials for $\mathbb{P}^k$ by $\mathcal{O}_{\mathbb{P}^k}(n)(\mathbb{P}^k)$.

Now, recall that a n-dimensional vector bundle $\pi : E \to X$ over a field K can be defined by taking open sets $U_i \subseteq X$ such that $\pi^{-1}(U_i) \to U_i \times F$ and insuring that for any intersection $U_i \cap U_j$ we have a gluing map:

$$g_{ij} : \pi^{-1}(U_i \cap U_j) \to \mathrm{GL}_n(K)$$

such that $g_{jk}g_{ij} = g_{ik}$ (the cocycle condition) which specifies the gluing condition. When limiting to line bundles, the gluing map becomes:

$$g_{ij} : \pi^{-1}(U_i \cap U_j) \to K^*$$

Intuitively, this give a point-wise gluing conditions on the bundle: if $v \in U_i$ and $w \in U_j$ that are the same point in $U_i \cap U_j$, then:

$$v = A_{ij}w$$

Moving onto the construction we have done earlier, as we are saying that $\mathcal{O}_{\mathbb{P}^1}(n)$ ought to be considered as a line bundle, we see that we ought to consider:

$$g_{ij} : \mathcal{O}_{\mathbb{P}^1}(U_i \cap U_j) \to \mathcal{O}^*_{\mathbb{P}^1}(U_i \cap U_j) \cong k[x, 1/y] \setminus \{0\}$$

where we take $\mathcal{O}^*_{\mathbb{P}^1}(U_i \cap U_j)$ in place of K^* as these are the "units" (i.e. isomorphisms) on this intersection. For example, given f, g restrict to the same functions on the intersection, then we now "twist" g to be:

$$f(x) = x^n g(1/x) \qquad x^n \in \mathcal{O}^*_{\mathbb{P}^1}(U_i \cap U_j)$$

The word "twist" comes from both complex analysis where z^n twists $\mathbb{C}$ onto itself around the branching point, and from algebraic geometry where the non-triviality of bundles can be interpreted by characteristic classes as some non-trivial twisting (we shall cover this intuition in ref:HERE). The details of this correspondence is given in [1, section 4.1] (especially how to transition from points of a bundle to section of a sheaf). Overall, we see that $\mathcal{O}_{\mathbb{P}^1}(n)(\mathbb{P}^1)$ is the global sections of a line bundle; more generally $\mathcal{O}_{\mathbb{P}^1}(n)$ remember all the sections over each open set U, as $\mathcal{O}_{\mathbb{P}^1}(n)$ is a sheaf.

With this intuition, we can see how to interpret $\mathcal{O}_X(D)$ as a line bundle. Choosing an affine cover U_i for X, we can define:

$$D|_{U_i} = \sum_{p \in U_i} n_p p$$

which is the representation of the divisor D in U_i. Then for each affine U_i, choose $s_i \in \mathcal{M}_X(U_i)$ so that $\langle s_i \rangle_K = \mathcal{O}_X(D)(U_i)$ is a generator as a K-algebra of the local sections and:

$$\operatorname{div}(s_i) = -D|_{U_i}$$

So that s_i has roots/poles of the appropriates order. Such a section exists, for $U_i \cong \mathbb{A}^1$ (as X is a curve) so $\mathcal{O}_X(U_i) \cong k[x]$ and so by our earlier discussion we can certainly find an s_i with exactly zeros and poles at $-D|_{U_i}$ and it is an exercise in ring-theory to show this is a generator (ex. think of choosing a uniformization parameter for a local ring).

Let us label the collection of the meromorphic functions $\mathcal{O}_X(D)(U_i)$. To find an $f \in \mathcal{O}_X(D)(X)$ (i.e. a global section), we need to insure we can glue together nicely on intersection. Consider (nonempty) $U_i \cap U_j$ and take the generators $s_i \in \mathcal{O}_X(D)(U_i)$ and $s_j \in \mathcal{O}_X(D)(U_j)$. It is not hard to see that $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$. Then certainly on $U_i \cap U_j$:

$$\operatorname{div}(s_i) = \operatorname{div}(s_j) = -D|_{U_i \cap U_j}$$

and so, as we are considering meromorphic functions, we can consider:

$$\operatorname{div}\left(\frac{s_j}{s_i}\right) = \operatorname{div}(s_j) - \operatorname{div}(s_i) = 0$$

As s_j/s_i has no zero's/pole on $U_i \cap U_j$, this means that it is holomorphic and nowhere vanishing on $U_i \cap U_j$. But then:

$$g_{ij} = \frac{s_j}{s_i} \in \mathcal{O}^*_X(U_i \cap U_j)$$

But then we have the desired transition functions! Thus, we may glue them to form all global sections $\mathcal{O}_X(D)(X)$for every D, and then $\mathcal{O}_X(D)$ is a line bundle. In most cases (and in most spaces that will be considered in this paper), every line bundle $\mathcal{L}$ is induced by a divisor D; [1, section 4.3] shows how there is a bijective correspondence between line bundles and (Weil/Cartier) divisors on the appropriate type of scheme.

1.2 Motivating the Generalization

here

1.2.1 Motivating Euler Characteristic

The origins of the Euler characteristic stem from an observation by Leonhard Euler about the relation between the vertices, edges, and faces of a convex Polyhedron. If V, E, F represents the respective quantities, then:

$$V - E + F = 2$$

For example, a cube has 8 vertices, 12 edges, and 6 faces, so $8 - 12 + 6 = 2$. Similarly, a closed connected planar graph holds the same property (counting the outside of the planar graph as a face). This sum is called the *Euler Characteristic* of the polyhedron. If the polyhedron was not convex, this relation need not hold, however it is possible to have a non-convex polyhedron whose Euler characteristic is 2. This shows that the Euler characteristic capture some rough information about the geometry of the shape.

Generalizing this, we can form CW-complex C on a topological space X and define the Euler characteristic as:

$$\chi(X, C) = \sum_{i=0}^{\infty} (-1)^i k_i$$

One of the crowning results of algebraic topology in the 21st century is the invariance of this number of choice of CW complex. This lead to the development of *singular cohomology* with the homology group $H_i(X, \mathbb{Z})$ and the *Betti numbers* $b_i = \text{rk}(H_i(X, \mathbb{Z}))$ and:

$$\chi(X) = \sum_{i=0}^{\infty} (-1)^i b_i = \sum_{i=0}^{\infty} (-1)^i \text{rk}(H_i(X, \mathbb{Z}))$$

As this is a homological constant, it is often called the *Euler-Poincaré* characteristic to distinguish it from the Euler characteristic for polyhedron, but we shall rarely make this distinction. As cohomology groups are invariant up to homotopy on X, the Euler characteristic is invariant to homotopy. It is often computationally easier to work with the cohomology group, and by the Poincaré duality (or its generalization, Serre duality), we have the equivalent sum: check no sign problem

$$\chi(X) = \sum_{i=0}^{\infty} (-1)^i \text{rk}(H^i(X, \mathbb{Z}))$$

How should we interpret the alternating nature of the sum when summing over homology/cohomology? The key idea is to notice that the alternating sum mimics the pattern when invoking the inclusion-exclusion principle. Looking at the Euler characteristic for polyhedron, we can take for example a cube, and hen we can think of each face as including the vertices and edges, and the edges as including the vertices. By counting all the faces, we have in fact double counted the edges. We thus subtract the sum of the edges. But this shall at the same time remove all the vertices from the sum, and hence we add them back in. This is exactly the inclusion-exclusion principle.

When converting to a cohomological perspective, we recover the inclusion-exclusion principle when taking *Čech cohomology*. In particular, the intuition is to cover your space X in open sets, define the desired type of functions on each open set, and measure the obstruction to gluing these functions on intersections, intersection of intersections, and so on and so forth:

img here

This same idea applies when considering the cohomology groups in the Euler characteristic sum.

How should we interpret what invariance is captured by the Euler characteristic? The case of the convex and non-convex polyhedron hints at a notion of curvature, and indeed by the Gauss-Bonnet theorem a Riemann surface X (without boundary) with Riemann metric ω which induces a Gaussian curvature K can be computer by taking:

$$\chi(M) = \frac{1}{2\pi}\int_M K dA$$

and generalizes to higher-dimensional surfaces (as shall be remarked on in section 1.2.2). These are very useful invariants, especially in 2 dimension where the classification of Riemann surfaces up to homotopy if first broken down into the case where the Euler characteristic is positive, 0, or negative (representing spherical, flat, and hyperbolic curvature respectively). However, we see that the Euler characteristic can be defined on any space admitting a cohomology theory, suggesting some more broad interpretation is applicable.

1.2.2 Motivating Characteristic Theory

By the discussion in sect ref:HERE, $\deg(D)$ and $1-g$ should be interpreted as a property of a bundle. Let us return to the Gauss-Bonnet Theorem to find the motivation of this generalization. Recall that if M is a compact two-dimensional Riemannian manifold with empty and K is be the Gaussian curvature of M, then:

$$\chi(M) = \frac{1}{2\pi}\int_M K dA$$

It was proven by Chern (the *Chern Theorem* or *Chern-Gauss-Bonnet Theorem*) that this generalizes to an even dimensional closed Riemann manifold to the curvature class of the Euler form:

$$\chi(M) = \int_M e(\Omega)$$

where e is the *Euler class*, and Ω is the 2-form representing curvature which can be embedded int:

$$\bigwedge^{2n} T^*M$$

Or more generally, if M is any smooth orientable n-dimensional manifold then:

$$\chi(M) = (e(TM), [M])$$

where $[M]$ is the fundamental class and $(-,-)$ is the cap product. The Euler class is a special form of the top Chern class.

A ith Chern class can be thought of as representing an obstruction to forming $n-i$ global linearly independent frames.

In a sense, this is the "geometric" part of the Riemann Roch theorem. If we let $D = 0$ so that we looking at the holomorphic functions, then:

$$\chi(M) = 1 - g$$

In terms of the Chern class, we get:

$$\chi(M) = \frac{1}{2}c_1(TM)$$

So something is going on here that is connected them but also non-trivially. (Important to note that there are two bundles going on: the “chosen” bundle $\mathcal{L}$ and the tangent bundle TM representing the geometric information).

then go over the formula in two dimensions to show that it really is non-trivial

So the goal now is to find a connection between the Euler-characteristic and some equation of Chern classes. We cannot directly use the total Chern class as χ and c are algebraically different, in particular if we define the *Chern character* denoted ch to be a representative power series of the Chern classes, then: fix this:

$$\mathrm{ch}(E_1 \oplus E_2) = \mathrm{ch}(E_1) + \mathrm{ch}(E_2) \qquad \chi(E_1 \oplus E_2) = \chi(E_1) \cdot \chi(E_2)$$

This shows that the Chern character is not the right invariant using Chern classes, but the Chern theorem suggests that the appropriate invariant does exist. It was *John Arthur Todd* who found this geomeric idea before Chern classes where even discovered (https://en.wikipedia.org/wiki/Todd_class) and *Hirzebruch* that put them to good use connecting them to find the “algebraic” information that would be stored in a bundle. In particular, we shall define the *Todd Character* which shall have the property on line bundles that:

$$\mathrm{td}(E_1 \oplus E_2) = \mathrm{td}(E_1) \cdot \mathrm{td}(E_2)$$

mimicking the Euler characteristic. Then we shall take advantage of the *splitting principle* as well as a “correction term” given by the Chern character ch to take into account the algebraic information to finally come up with:

$$T(M, E)) = \kappa_n(\mathrm{ch}(E)\mathrm{td}(TM))[M] = \int_X \mathrm{ch}(X)\mathrm{td}(X)$$

where κ_n is the evaluation at the fundamental class, and the right hand side is the same evaluation except using the integration notation due to the isomorphisms with De-Rham cohomology groups. Notice how this can be seen as a great generalization of Chow’s Theorem, where we are not limited to just looking at the curvature form, but had to limit to smooth varieties in order for the “data” to be well-defined (and this can be pushed quite far as shown in ref:HERE(the new paper from 21st century) This shall finally give us:

$$\chi(M, E) = T(M, E)$$

which is our desired result.

should definitely mention the connection of Chern character to K-theory, the homomorphism, the natural transformation form K-theory to cohomology, why the diagram doesn’t commute, and how it is fixed with the Todd character.

2

GAGA results

Throughout this paper, we shall often freely interchange result on compact complex manifolds (often Kähler manifolds) with results on projective varieties. In order for this to be done without issue, the important results from *Géométrie algébrique et géométrie analytique* (or GAGA) must be outlined. We shall follow more closely ref:HERE.

Define Z-topology and $\mathbb{C}$-topology on a variety/closed manifold X.

(Part of the point of this is that we will want to use things like para-compactness that gives bijective correspondence between some cohomology groups, but partitions of unity do not exist on Z-topology, so we “fix this” by the GAGA result)

3

Reviewing Characteristic Classes

I think I need a word here on how to also translate all of this to sheaves!!

Let $E \to X$ be vector bundle, where X is paracompact and finite dimensional (either as a manifold or topologically)[1]. A natural set of questions to ask is what are the possible (global) sections on E with respect to X. A classical example of this question is to ask if there are any non-vanishing global sections on TS^2 with respect to S^2; *Hedgehog Theorem* (or infamously, the *Hairy Ball Theorem*) asserts that no such global section can exist [6]. As local sections are easy to produce by taking advantage of local trivialization, the question is what type of *obstructions* are there in producing a global sections. This is thus a cohomological problem.

By the work of mathematicians such as Chern, Thom, Whitney, and many others, there are natural transformations from the Vector-bundle functor to the cohomology functor of the underlying space:

$$c : K_k(-) \Rightarrow H^*$$

where $K_k(X)$ is the isomorphism classes of k-vector bundles over X, and k is a field (usually $\mathbb{R}$ or $\mathbb{C}$). Without much trouble, this can be generalized to principal G-bundles; this shall be done for the important case of Flag bundles and Grassmannians that will allow for both classification results and computational results. In this case, a *characteristic class* would be a natural transformation:

$$c : b_G(-) \Rightarrow H^*$$

where $b_G(-)$ is the functor taking X to the isomorphism of principal G-bundles over X. Note that there can be more than one characteristic class. Uncurling the categorical language, these natural transformations associate to each principal G bundle $P \to X \in b_G(X)$ an element $c(P) \in H^*$ such that $c(f^*(P)) = f^*(c(P))$ (i.e. the pullback of the vector bundle commutes with the pullback of cohomology classes) that is true up to isomorphism (and even homotopy). The characteristic classes

[1]these conditions insure we get bijective maps between the cohomological spaces of sheaves and the underlying space

will be the "right" notion to capture the cohomological information, in particular when working over real and complex vector bundles, the cohomology of the classifying space (MO for real, MU for complex) is given by the *Stiefel-Whitney* and *Chern* classes (see [5]):

$$H^*(MO; \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}_2[w_1, ..., w_n] \qquad \text{and} \qquad H^*(MU; \mathbb{Z}) \cong \mathbb{Z}[c_1, ..., c_n] \tag{3.1}$$

As our base spaces are (complete) varieties, we shall mainly focus on complex vector bundles and principal $\mathrm{GL}_n(\mathbb{C})$-bundles. The complex case works with *Chern class* and the *Pontryagin class*: the Chern class is the natural characteristic class of complex vector bundles (as seen in equation (3.1)), while the Pontryagin class can be thought of as the characteristic classes of the complexification of a real vector bundle.

The Chern class is central to the Hirzebruch Riemann Roch as they are directly used in defining the Chern character and the Todd character. On the other hand, the Pontryagin classes shall be useful in that they will allow us to fully classify all cobordant rings, a result that will be necessary for only requiring checking χ agree s with $\int_X \mathrm{ch}(E)\mathrm{td}(T_X)$ on complex projective space.

Another characteristic class that shall not cover be covered but is important to know as it is one of the generalizations of the Euler characteristic is the *Euler class*. As the name suggests, the Euler class is linked to the Euler characteristic, in particular if M is an oriented closed n-manifold, TM the tangent bundle, $e(TM) \in H^n(M; \mathbb{Z})$, and $[M] \in H_n(M; \mathbb{Z})$ the fundamental class of M, then:

$$e(TM) \frown [M] = \chi(M)$$

or, given the DeRham isomorphism, re-labeling the notation $e(TM)$ to the corresponding *Euler form*:

$$\int_M e(TM) = \chi(M)$$

If M where furthermore a Riemann manifold, then the classical generalization of the Gauss-Bonnet theorem for manifolds without boundary (the *Chern Theorem*) let's us replace e with Ω, the curvature form, giving:

$$\int_M e(\Omega) = \chi(M)$$

In other words, the Chern Theorem finds an alternative way of evaluating the Euler class at the fundamental class that relies on the curvature form Ω when it exists. The Hirzebruch Riemann Roch can be seen as finding an alternative evaluation at the Euler class when the underlying space M has enough structure for its characterization to be done via the Chern classes.

Finally, the definition of Chern class and Pontryagin class taken here can be thought of as the "classical" definition. In algebraic geometry, there is natural "intersection theory" perspective of of the Chern class that maps the chow group $A^n(-)$ to $A^{n-k}(-)$. There is a natural way in which this definition is linked to the classical definition of Chern class via the Chern character which shall be covered in ref:HERE, and the we shall briefly bring it up in the context of the generalization of the Hirzebruch Riemann Roch to the *Grothendeick Riemann Roch theorem* (see ref:HERE) to show an important representation of the $K_0(X)$ group using chow groups.

3.1 Definitions and key Results

For reference, here is the definition of a vector bundle and Principal bundle we shall use (this differs from Hirzebruch gives the gluing definition of a vector bundle with structure groups):

Definition 3.1.1: Fiber Bundle

Let B be a paracompact topological space. Then a quadruple (B, E, π, F) where E is a topological space called the *total space*, $\pi : E \to B$ is a continuous surjection called the *bundle projection*, and B is called the *base space*, and F is the fiber is called *fiber bundle* if:

1. (regular fibers) each $\pi^{-1}(b)$ for $b \in B$ is isomorphic to F, or in particular

$$\pi^{-1}(b) \cong b \times F$$

2. (local trivialization)For each $b \in B$, there exists an open set $U \subseteq B$ containing b such that the following diagram commutes

$$\begin{tikzcd}
\pi^{-1}(U) \arrow[r, "\sim"] \arrow[d, "\pi"'] & U \times F \arrow[dl, "\pi_1"] \\
U &
\end{tikzcd}$$

When clear from context, the fiber bundle is usually denoted as E.

Definition 3.1.2: G-bundle

Let $E \to X$ be a fiber bundle and G a topological that acts faithfully on the fibers. Then a *G-atlas* is a collection of local trivializations $\{U_i, \varphi_i\}_i$ such that for any index i, j on the overlap $U_i \cap U_j$ (if it is non-empty):

$$\varphi_j^{-1} \circ \varphi_i : (U_i \cap U_j) \times F \to (U_i \cap U_j) \times F$$
$$\varphi_j^{-1} \circ \varphi_i(p, f) = (p, g_{ij}(p)f)$$

with the added conditions that:

1. $g_{11} = \text{id}$
2. $g_{ij} = g_{ji}^{-1}$
3. $g_{ik} = g_{jk}g_{ij}$

Two G-atlases are equivalent if their union is a G-atlas. A *G-bundle* is a fiber bundle with an equivalence relation of G-atlases.

Definition 3.1.3: Principal G-Bundle

If $F = G$ is a topological and $P \to X$ is a fiber bundle over X group where there is a free and transitive continuous right group action $P \times G \to P$ that preserves the fibers of P, then it is called a *principal G-bundle* (or *principal bundle* if G is clear from context. In other words, a principal G-bundle is a G-bundle where the fibers are G.

Each fiber is called a *G-torsor* (which is a space that where the action $G \curvearrowright F_x$ has trivial stabilizer, hence "is" the group G but does not intrinsically have a notion of identity element)

Definition 3.1.4: Vector Bundle

If $F = V$ is a vector space and each fiber-morphism in the local trivialization is a linear isomorphism, then is called a *vector bundle* (over B). If the vector space is real or complex, the vector bundle can be called a *real vector bundle complex vector bundle*.

We shall also freely use the following fact (proven in [5]):

Theorem 3.1.5: Homotopy Invariance Vector Bundles

Let $p : E \to B$ a vector bundle and $f_0, f_1 : A \to B$ be homotopic maps. Then if A is paracompact,

$$f_0^*(E) \cong f_1^*(E)$$

Proof **:**
[5, section 1.2]

Theorem 3.1.6: Classifying Vector Bundles

Let X be a paracompact space. Then $K^n(X)$ is in correspondence with the set of homotopy classes of maps $X \to \mathrm{Grass}_n$, denoted $[X, \mathrm{Grass}_n]$. In other words:

$$K^n(X) \cong [X, \mathrm{Grass}_n]$$

In particular, if $E \to X$ is an n-dimensional vector bundle, it is isomorphic to $f^*(E_n)$ for some $f : X \to \mathrm{Grass}_n$ where E determines f up to homotopy.

Proof **:**
[5, section 1.2]

This motivates the following natural definition:

Definition 3.1.7: Universal Bundle

The Grassmannians $\mathrm{Grass}_{n,k}$ are called the *universal bundles*; for $\mathbb{R}, \mathbb{C}, \mathbb{H}$ they are often denoted BO_n, BU_n,and BSp_n respectively.

3.2 The splitting Principle

One the most important results about vector bundles is that there is a pullback map which allows them to split into a direct sum of line bundles. As a consequence, the total Chern class (which is representable as a polynomial) can be split into product of Chern classes of line bundles (which can be represented by linear terms). The key cohomological result is the following "fiber-local to global" theorem proved by Leray and Hirsch:

Theorem 3.2.1: Leray-Hirsch Theorem

Let R be a commutative ring and $p : E \to X$ be a fiber bundle. In [6] it was shown $H^*(E; R)$ is a $H^*(X; R)$-module where

$$\alpha \cdot \beta := p^*(\alpha) \smile \beta$$

with $\alpha \in H^*(X)$ and $\beta \in H^*(E)$. If

1. for each fiber F the inclusion $F \hookrightarrow E$ induces a surjection on $H^*(-; R)$
2. $H^n(F; R)$ is a free R-module of finite rank for each n.

Then $H^*(E; R)$ is a free $H^*(X; R)$-module. A basis for the $H^*(X; R)$-module $H^*(E; R)$ can be obtained by choosing any set of elements in $H^*(E; R)$ that map to a basis for $H^*(F; R)$ under the map induced by the inclusion.

***Proof* :**
see [6].

Theorem 3.2.2: Splitting Principle

Let $\pi : E \to B$ be a complex vector bundle. Then there exists a space $F(E)$ and a map $p : F(E) \to B$ such that the pullback $p^*(E) \to F(E)$ splits as a direct sum of line bundles, and

$$p^* : H^*(B; \mathbb{Z}) \hookrightarrow H^*(F(E); \mathbb{Z})$$

is injective

***Proof* :**
Consider the pullback $\mathbb{P}(\pi)^*(E)$ of E via the map $\mathbb{P}(\pi) : \mathbb{P}(E) \to B$ where $\mathbb{P}(E)$ is the space of lines through the origin in all fibers of E. This pullback contains a natural one-dimensional subbundle

$$L = \{(\ell, v) \in \mathbb{P}(E) \times E \mid v \in \ell\}$$

An inner product on E pulls back to an inner product on the pullback bundle, and thus a splitting of the pullback as a sum $L \oplus L^{\perp}$ with the orthogonal bundle $L^{\perp}$ having dimension one less than E. Then by the Leray-Hirsch theorem to $\mathbb{P}(E) \to B$, $H^*(\mathbb{P}(E);\mathbb{Z})$ is the free $H^*(B;\mathbb{Z})$-module with basis $1, x, \ldots, x^{n-1}$ and in particular the induced map $H^*(B;\mathbb{Z}) \to H^*(\mathbb{P}(E);\mathbb{Z})$ is injective since one of the basis elements is 1.

This construction can be repeated with $L^{\perp} \to \mathbb{P}(E)$ in place of $E \to B$, until we get the splitting, completing the proof.

3.3 Chern and Pontryagin Classes

We take the axiomatic approach taken by Grothendeick:

Theorem 3.3.1: Existence and Uniqueness of Chern Classes

Let $E \to X$ be a complex vector bundle. Then there exists a sequence of functions $c_1, c_2, ...$ where $c_i(E) \in H^{2i}(X;\mathbb{Z})$ that depends only on the isomorphism type of E such that

1. $c_i(f^*(E) = f^*(c_i(E))$ for a pullback $f^*(E)$
2. $c(E_1 \oplus E_2) = c(E_1) \smile c(E_2)$ where $c = 1 + c_1 + c_2 + \cdots \in H^*(X;\mathbb{Z})$
3. $c_i(E) = 0$ if $i > \dim E$
4. Given $E \to \mathbb{C}\mathrm{P}^\infty$ is the canonical line bundle, $c_1(E)$ is a generator of $H^1(\mathbb{C}\mathrm{P}^\infty;\mathbb{Z})$ specified in advance.

Proof **:**
see [5] for a full proof. The key idea is to invoke the Leray-Hirsch Theorem for existence and the splitting principle for uniqueness. In fact, we get a bit more out of the Leray-Hirsch theorem: by the classification of vector bundles (theorem 3.1.6) there is map $f : E \to \mathbb{C}\mathrm{P}^\infty$ which induces a cohomology pullback $f^* : H^*(\mathbb{C}\mathrm{P}^\infty;\mathbb{Z}) \to H^*(\mathbb{P}(E);\mathbb{Z})$, in particular $f^* : H^1(\mathbb{C}\mathrm{P}^\infty;\mathbb{Z}) \to H^1(\mathbb{P}(E);\mathbb{Z})$ which pullback the generator $\alpha \in H^1(\mathbb{C}\mathrm{P}^\infty;\mathbb{Z})$ to an element $f^*(\alpha) = x$. Then the elements $1, x, ..., x^n$ are the generators of $H^*(\mathbb{P}(E),\mathbb{Z})$ as a free $H^*(X;\mathbb{Z})$-module, and if $\alpha\beta := \alpha \smile \beta$ then we get a unique relationship:

$$x^n - c_1(E) + \cdots (-1)^n c_n(E) \cdot 1 = 0$$

which gives a *relationship* between the Chern classes.

Uniqueness is given by the splitting principle, namely if here was another such relationship, then as the above equation must pullback and split, the Chern classes must match.

For the choice of normalization, note that for property (4), the canonical line bundle over $\mathbb{C}\mathrm{P}^1$ could have been used instead of $\mathbb{C}\mathrm{P}^\infty$ as $\mathbb{C}\mathrm{P}^1 \hookrightarrow \mathbb{C}\mathrm{P}^\infty$ induces an isomorphism

$$H^1(\mathbb{C}\mathrm{P}^\infty;\mathbb{Z}) \cong H^1(\mathbb{C}\mathrm{P}^1;\mathbb{Z})$$

and we know by some cohomology theory that we can choose any hyperplane for the fundamental class of $H^1(\mathbb{C}\mathrm{P}^1;\mathbb{Z})$, hence $c_1(E) \in H^1(\mathbb{C}\mathrm{P}^1;\mathbb{Z})$ becomes a choice of hyperplane.

Note that this definition differs slightly from what Hirzebruch presents: he replaces the third axiom with the following:

(3) If $\xi_1, \ldots, \xi_q$ are continuous $\mathbf{U}(1)$*-bundles over X then*

$$c(\xi_1 \oplus \cdots \oplus \xi_q) = c(\xi_1) \smile \cdots \smile c(\xi_q).$$

which is equivalent due to the splitting principle. The Chern classes are the "natural" choice of

theorem of classifying cohomology of universal bundle

The real equivalent has more subtly due to the fact that real manifolds and vector bundles have a choice of orientation, and hence the natural cohomological setting to work in is over $\mathbb{Z}/2\mathbb{Z}$-coefficient and classify $H^*(BO; \mathbb{Z}/2\mathbb{Z})$ where we take $\mathbb{Z}/2\mathbb{Z}$ to get rid of choice of orientability. This uses *Stiefel-Whitney* classes which we did not touch on in this paper. To classify $H^*(BO; \mathbb{Z})$, we essentially take the "complexification" of a real vector bundle, which gives us the following best possible result:

Definition 3.3.2: Pontryagin Class

Let $E \to B$ be a real vector bundle. Complexity it by adding the almost complex structure to $E \oplus E$:

$$E^{\mathbb{C}} = E \oplus E \qquad i(x, y) = (-y, x)$$

Then a *Pontryagin class* $p_i(E) \in H^{4i}(B; \mathbb{Z})$ is given by:

$$p_i(E) = (-1)^i c_{2i}(E^{\mathbb{C}}) \in h^{4i}(B; \mathbb{Z})$$

To characterize Pontryagin classes, we state the following:

Proposition 3.3.3: Characterizing Grassmannian Cohomology

All torsion of $H^*(BO, \mathbb{Z})$ is over 2; if T is the torsion subgroup then:

$$\frac{H^*(BO, \mathbb{Z})}{T} \cong \mathbb{Z}[p_1, ..., p_k] \qquad n = 2k \text{ or } 2k+1$$

Proof **:**
This is not important for our current work, so the proof is deferred to [5] or [16]

What shall be important to use is the use of Pontryagin classes in the theory of cobordism in section 5.

Should put here the theorem linking Pontryagin and Chern class, ex. Theorem 4.6.1 in Hirzebruch or an equivalent from Hatcher.

3.4 Chern Character and K-Theory

Fix a base space X. Let $c(-) : K(X) \to H^*(X; \mathbb{Z})$ be the Chern class. Recall that

$$c(E_1 \oplus E_2) = c(E_1) \smile c(E_2)$$

In particular, it brings the addition in $K(X)$ to multiplication in H^*. The collection $K(X)$ can be completed to form a group, also denoted $K(X)$, and it is known as the *Grothendieck K-group*. The tensor gives it a natural ring structure. By the above construction, we see that $c(-)$ is *not* a ring homomorphism. From the Chern class $c(-)$, we can construct a new map known as the *Chern character*

$$\mathrm{ch}(-) : K(X) \to H^*(X;\mathbb{Z}) \otimes \mathbb{Q}$$

which *will* be a ring homomorphism, provided we add rational coefficients.

By the splitting principle, it suffices to work with line bundles. The total Chern class is:

$$c(L) = 1 + c_1(L)$$

where $c_1(L) \in H^2(X;\mathbb{Z})$. As shown in ref:HERE,

$$c_1(L_1 \otimes L_2) = c_1(L) + c_1(L)$$

which as addition. To fix this into multiplication, we can take:

$$\mathrm{ch}(L) = e^{c_1(L)} = 1 + c_1(L) + \frac{c_1(L)^2}{2!} + \frac{c_1(L)^3}{3!} + \cdots$$

where the power should be taken as repeated cup product. This fixes the multiplicative problem for:

$$\mathrm{ch}(L_1 \otimes L_2) = e^{c_1(L_1)+c_1(L_2)} = e^{c_1(L_1)} \cdot e^{c_1(L_2)}$$

But this introduces rational coefficients. As $H^*(X;\mathbb{Z})$ can have torsion, ex. $H^*(\mathbb{C}\mathrm{P}^1;\mathbb{Z}) \cong \mathbb{Z}[h]/(h^2)$ (see [6, appendix]), many of the terms here disappear and it is not well-defined. Thus, we must define:

$$H^*(X;\mathbb{Q}) := H^*(X;\mathbb{Z}) \otimes \mathbb{Q}$$

which in [14] it is shown that the torsion is eliminated, and hence $\mathrm{ch}(L) \in h^*(X;\mathbb{Q})$ is well-defined. Then, for an arbitrary vector bundle E, we can find a unique splitting

$$E = L_1 \oplus L_2 \oplus \cdots \oplus L_n$$

and define:

$$\mathrm{ch}(E) = \mathrm{ch}(L_1) + \mathrm{ch}(L_2) + \cdots + \mathrm{ch}(L_n)$$

where each $\mathrm{ch}(L_i) = t_i$ are the *Chern root* of $c(E)$. By construction:

$$\mathrm{ch}(E_1 \oplus E_2) = \mathrm{ch}(E_1) + \mathrm{ch}(E_2)$$

Giving the desired algebraic properties. However, $\mathrm{ch}(E)$ is dependent on the Chern classes $c_1(L_i)$ where i goes over the Chern roots, as opposed to $c_j(E)$ where j increased in dimension. This is easily fixed with a bit of combinatorics. Namely, $c_j(E)$ is given by:

$$c_j(E) = \sigma_k(c_1(L_1), ..., c_1(L_n))$$

where σ_k is the kth elementary-symmetric polynomial. Define the *power sums* to be:

$$s_k(x_1, ..., x_n) = \sum_i^n x_i^k$$

Then the *Newton identities* (see this link) show that (via elementary computations):

$$\begin{aligned}
s_1 &= \sigma_1 \\
s_2 &= \sigma_1^2 - 2\sigma_2 \\
s_3 &= \sigma_1^3 - 3\sigma_2\sigma_1 + 3\sigma_3 \\
s_3 &= e_1^4 - 4e_2e_1^2 + 4e_3e_1 + 2e_2^2 - 4e_4 \\
&\vdots
\end{aligned}$$

From these we get:

$$\mathrm{ch}(E) = \sum_{i=1}^{n} \left(\sum_{k=0}^{\infty} \frac{t_i^k}{k!} \right) = \sum_{k=0}^{\infty} \frac{1}{k!} \left(\sum_{i=1}^{n} s_k(c_1(E), ..., c_n(E)) \right)$$

Writing out the first few terms we get:

$$\mathrm{ch}(-) = \dim_{\mathbb{C}}(-) + c_1 + \frac{1}{2}(c_1^2 - 2c_2) + \frac{1}{6}(c_1^3 - 3c_1c_2 + 3c_3) + \cdots$$

giving us a power series purely in terms of the Chern classes of E! Note that the Chern character in facts captures more information than the Chern classes as the Chern classes are *stable*, meaning they are invariant when taking the direct sum with trivial bundles, however $\mathrm{ch}(-)$ has $\dim_{\mathbb{C}}(-)$ in its' 0th position, in particular it remembers the rank of the bundle[2] To connect this to the K-group, we require the following:

Proposition 3.4.1: Extension and Tensors With Chern Character

let ξ'', ξ, ξ' be vector bundles such that:

$$0 \to \xi'' \to \xi \to \xi' \to 0$$

then:

$$\mathrm{ch}(\xi)(t) = \mathrm{ch}(\xi')(t) + \mathrm{ch}(\xi'')(t)$$

And similarly

$$\mathrm{ch}(\xi \otimes \eta)(t) = \sum_{i,j} e^{(\gamma_j + \delta_j)t} = \sum_{i,j} e^{\gamma_j t} e^{\delta_j t} = \left(\sum_i e^{\gamma_j t} \right) \left(\sum_j e^{\gamma_j t} \right) = \mathrm{ch}(\xi)(t)\,\mathrm{ch}(\eta)(t).$$

***Proof* :**
exercise.

$$\mathrm{ch}(E_1 \otimes E_2) = \mathrm{ch}(E_1) \cdot \mathrm{ch}(E_2)$$

where again $\cdot$ is the cup product. Thus, there is a natural map:

$$\mathrm{ch} : K(X) \to H^*(X; \mathbb{Q})$$

[2] As a fun aside, if $X = \mathbb{C}\mathrm{P}^n$, then the Hilbert polynomial can be recovered by the Chern character, and vice-versa; see [3]

This shows that ch is the right notion for capturing the cohomological information of vector bundles which respects the algebraic structure of $K(X)$. In chapter 6, we shall show that $\mathrm{ch}(E)$ capture the bundle information, while $\mathrm{td}(X)$ (define in chapter 6) shall capture the relevant information from the underlying space.

A natural question to ask is whether $f : X \to Y$ is a continuous map, then ch commutes with f^*, namely if:

$$\begin{tikzcd} K(Y) \arrow[r, "f^*"] \arrow[d, "\mathrm{ch}"'] & K(X) \arrow[d, "\mathrm{ch}"] \\ H^*(Y;\mathbb{Q}) \arrow[r, "f^*"] & H^*(X;\mathbb{Q}) \end{tikzcd}$$

However this is not the case. What fixes the commutativity is the introduction of the Todd class. It is in fact a little more complicated as we would usually be taking coherent sheaves and working with the pushforward rather than the pullback, and so the details are left for section ref:HERE.

$$\begin{tikzcd} K(X) \arrow[r, "f_!"] \arrow[d, "\mathrm{ch}(-)\mathrm{td}(TX)"'] & K(Y) \arrow[d, "\mathrm{ch}(-)\mathrm{td}(TY)"] \\ H^*(X;\mathbb{Q}) \arrow[r, "f_*"] & H^*(Y;\mathbb{Q}) \end{tikzcd}$$

4

Cohomology of Varieties – new Euler Characteristics

The Euler characteristic is in a sense "too rough" of an invariant for varieties. We require to capture some of the complex structure to find the right invariants. In particular, we shall refine the usual De-Rham/singular cohomology into *Dolbeault* cohomology which shall capture the obstruction to a function/local section being *holomorphic*. This shall form a new collection cohomology groups (depending on the "degree") that will allow us to define new Euler-characteristics that generalize the topological Euler Characteristic.

We start by developing some basic Hodge theory. The key point is to show that the Dolbeault cohomology groups are finite dimensional, which is achieved by using the fact that elliptic operators have finite-dimensional kernels (see ref:HERE). Note that in the proof of Grothendieck Riemann Roch, this is replaced with the fact that $H^i(X, \mathscr{F})$ is finite-dimensional for X projective and F coherent (see [1], or [19]).

Then, as all smooth projective manifold can be embedded into $\mathbb{C}\mathrm{P}^n$ have an algebraic equation by Chow's theorem (ref:HERE), there is a natural compatibility between the Riemann metric, symplectic form, and complex structure. This shall define Kähler manifolds, which are "almost" a variety; after defining what it means to be positive (essentially, insuring the tangent bundle is "very ample") we shall show that all positive Kähler manifolds are a (quasi-projective) smooth variety. This will let us translate the results over to algebraic geometry.

We then show how many interesting new Euler-characteristics can be defined. We shall then end off by mentioning some important Vanishing theorems that would allow us to compute $H^0(X, \mathcal{F})$.

Finally, we define the derived Euler-characteristic which shall keep track of important intersection information that will be important in the proof of Hirzebruch Riemann Roch.

4.1 Dolbeault Cohomology

Let X be a complex manifold. For now, let it be one-dimensional. As holomorphic mappings respects complex and anti-complex structure, any 1-form with complex coefficients can be uniquely with respect written as a sum:

$$f dz + g d\bar{z}$$

let $\Omega^{1,0}$ represent holomorphic 1-forms, and $\Omega^{0,1}$ represent anti-holomorphic 1-forms.The spaces $\Omega^{1,0}$ and $\Omega^{0,1}$ are both stable under holomorphic coordinate changes (use Cauchy-Riemann).

A similar pattern takes shape in higher dimensions: let X be n-dimensional. Like before, any 1 form with complex coefficients can be written as:

$$\sum_{j=1}^{n} (f_j dz^j + g_j d\bar{z}^j)$$

Define:

$$\Omega^{p,q} = \underbrace{\Omega^{1,0} \wedge \cdots \wedge \Omega^{1,0}}_{p \text{ times}} \wedge \underbrace{\Omega^{0,1} \wedge \cdots \wedge \Omega^{0,1}}_{q \text{ times}}$$

Then we know that for every $r \in \mathbb{N}$, every smooth r-form on X (with $\mathbb{C}$-coefficients) can be written as the sum of (p,q)-forms where $p + q = r$ (as the basis of these forms are wedge product of the lower dimensions, see [15], [6]).

Now, let E^k be the space of all complex differential forms of total degree k. Then

$$E^k = \Omega^{k,0} \oplus \Omega^{k-1,1} \oplus \cdots \oplus \Omega^{1,k-1} \oplus \Omega^{0,k} = \bigoplus_{p+q=k} \Omega^{p,q}$$

When the manifold is nice enough, ex. Kähler, $\Omega^k = E^k$ (where Ω^k is the collection of holomorphic forms, "decomposition" like we did above, that is the *Hodge decomposition*). Note that the direct sum decomposition is stable under holomorphic coordinate changes, hence it determines a vector bundle decomposition. Note we get a natural projection for each direct summand:

$$\pi^{p,q} : E^K \to \Omega^{p,q}$$

Now, define $d : \Omega^r \to \Omega^{r+1}$ in the natural way where:

$$d(\Omega^{p,q}) \subseteq \bigoplus_{r+s=p+q+1} \Omega^{r,s}$$

When we were just working with smooth manifolds, this gave us "enough" cohomological information. We now have more structure and this is not granular enough. We define the *Dolbeault operators*:

In coordinates, let:

$$\alpha = \sum_{|I|=p,|J|=q} f_{IJ} dz^I \wedge d\bar{z}^J \in \Omega^{p,q}$$

where I and J are multi-indices. Then:

$$\partial\alpha = \sum_{|I|,|J|} \sum_{\ell} \frac{\partial f_{IJ}}{\partial z^\ell} dz^\ell \wedge dz^I \wedge d\bar{z}^J$$

and

$$\bar{\partial}\alpha = \sum_{|I|,|J|}\sum_{\ell}\frac{\partial f_{IJ}}{\partial \bar{z}^{\ell}}d\bar{z}^{\ell}\wedge dz^I\wedge d\bar{z}^J.$$

Furthermore:

$$d = \partial + \bar{\partial}$$
$$\partial^2 = \bar{\partial}^2 = \partial\bar{\partial} + \bar{\partial}\partial = 0.$$

Thus, we see there is some interesting cohomology that can occur; this defines *Dolbeault cohomology.* The symbol ∂ suggests that they are dual to d, and indeed the Poincaré lemma generalizes to complex manifolds.

Let us define the Dolbeault cohomology. Notice that $\overline{\partial} f = 0$ if f is holomorphic, therefore it is in some sense the "natural" choice; the other choice keeping track of the "anti-holomorphic" information. Thus, define:

$$H^{p,q}(M) = \frac{\ker(\overline{\partial} : \Omega^{p,q} \to \Omega^{p,q+1})}{\operatorname{im}(\overline{\partial} : \Omega^{p,q-1} \to \Omega^{p,q})}$$

If we wanted to be precise, we could say $H^{p,q}_{\overline{\partial}}(M)$. As

$$H^{p,q}_{\overline{\partial}}(M) = \overline{H^{p,q}_{\partial}(M)}$$

We see that the difference of choice is a difference in conjugation, and hence we stick to $\overline{\partial}$ for the reasons mentioned before.

For our purposes, we will naturally want to generalize this to the Dolbeault cohomology of vector bundles. let E be a holomorphic vector bundle on a complex manifold X. Then take the map:

$$\overline{\partial}_E : \Gamma(E) \to \Omega^{0,1}(E)$$

taking section of E to $(0,1)$-forms with values in E. This will satisfy the Leibniz rule with respect to $\overline{\partial}$, and is thus a $(0,1)$-connecting on E. Then we can do the same trick from Riemann geometry and extended it to the exterior covariant derivative, namely;

$$\overline{\partial}_E : \Omega^{p,q}(E) \to \Omega^{p,q+1}(E)$$

which acts on on a section $\alpha \otimes s \in \Omega^{p,q}(E)$ by:

$$\overline{\partial}_E(\alpha\otimes s) = (\overline{\partial}\alpha)\otimes s + (-1)^{p+q}\alpha\wedge\overline{\partial}_E s$$

This naturally extends linearly to any section of $\Omega^{p,q}(E)$ and $\overline{\partial}_E^2 = 0$. Thus, the Dolbeault cohomology with coefficient in E is:

$$H^{p,q}(X,(E,\bar{\partial}_E)) = \frac{\ker(\bar{\partial}_E : \Omega^{p,q}(E) \to \Omega^{p,q+1}(E))}{\operatorname{im}(\bar{\partial}_E : \Omega^{p,q-1}(E) \to \Omega^{p,q}(E))}$$

It is standard to check that the Dolbeault cohomology groups do not depend on the choice of Dolbeault operator $\overline{\partial}_E$ compatible with the holomorphic structure of E, and so it is simply denoted by:

$$H^{p,q}(X,E)$$

phrase this in a better way for masters thesis There is a ton of fascinating theory here which I will leave for when I explore hodge theory. Let Ω^p be the sheaf of holomorphic p-forms on M. Then:

$$H^{p,q}(M) \cong H^q(M, \Omega^p)$$

and if we are working over a vector bundle:

$$H^{p,q}(M, E) \cong H^q(M, \Omega^p \otimes E)$$

This is *Dolbeault's Theorem*. This also means that:

$$H^{0,q}(M, E) \cong H^q(M, \Omega^0 \otimes E) = H^q(M, E)$$

For future reference, if E is a complex vector bundle, we shall denote its conjugate by $\overline{E}$ (defined in the natural way) which is anti-complex and anti-isomorphic to E. We can also define the dual bundle E^* in the natural way. Note that per-fiber, the dual can be given by Hermitian of the inverse of the matrix.

Hodge Theorem

Let us build-up to showing that each class in $H^{p,q}(M, \mathbb{C})$ is representable by a harmonic form. For notational simplicity, let:

$$A^{p,q}(W) = \Gamma(V, \Omega^{p,q}(W))$$

and:

$$A^{p,q} = A^{p,q}(\mathbb{1})$$

where $\mathbb{1}$ is the trivial bundle representing holomorphic functions. Let M now be compact. We shall define the Hodge operator. If W is a complex vector bundle, we can define W^* and a (perfect) pairing:

$$A^{p,q}(W) \times A^{r,s}(W^*) \to A^{p+r,q+s}(\mathbb{1}) \qquad (\alpha, \beta) \mapsto \alpha \wedge \beta$$

If $W = \mathbb{1}$, this is the usual wedge (or exterior) product. The product is anti-commutative in the sense that:

$$\alpha \wedge \beta = (-1)^{(p+r)(q+s)} \beta \wedge \alpha$$

Choosing $r = n - p$, $s = n - q$ we get the top form $\alpha \wedge \beta \in A^{n,n}(\mathbb{1})$. We can thus define:

$$\langle \alpha, \beta \rangle = \int_M \alpha \wedge \beta$$

By Stokes theorem, if $\alpha = \overline{\partial}\gamma$ for $\gamma \in A^{p,q-1}(W)$ and $\overline{\partial}\beta = 0$, (or if $\overline{\partial}\alpha = 0$ and $\beta = \overline{\partial}\gamma$) then:

$$\langle \alpha, \beta \rangle = \int_M \alpha \wedge \beta = \int_M \overline{\partial}(\gamma \wedge \beta) = \int_M d\,(\gamma \wedge \beta) = 0$$

Thus, we get a well-defined pairing:

$$H^{p,q}(M, E) \times H^{n-q,n-q}(M, E^*) \to \mathbb{C} \qquad (\alpha, \beta) \mapsto \langle \alpha, \beta \rangle = \int_M \alpha \wedge \beta$$

Now, by some linear algebra and differential geometry we see that $\overline{TM} \cong T^*M$. Then we get:

$$\begin{aligned}\bigwedge^p TM \otimes \bigwedge^q TM &\cong \bigwedge^p TM \otimes \bigwedge^n T^*M \otimes \bigwedge^n TM \otimes \bigwedge^q T^*M \\ &\cong \bigwedge^{n-p} T^*M \otimes \bigwedge^{n-q} TM \\ &\cong \bigwedge^{n-q} TM \otimes \bigwedge^{n-p} \overline{TM}\end{aligned}$$

There is thus a natural duality operator:

$$* : \bigwedge^p TM \otimes \bigwedge^q TM \to \bigwedge^{n-q} TM \otimes \bigwedge^{n-p} \overline{TM}$$

and $* \circ *$ becomes multiplication by $(-1)^{p+q}$.

Now, by vector-bundle theory we know we can without loss of generality reduce W to a unitary group bundle. There is then the natural hermitian anti-isomorphism:

$$\psi : W \to W^* \qquad \psi^{-1} : W^* \to W$$

If κ is the anti-isomorphism from $\bigwedge^r TM \otimes \overline{\bigwedge^s TM}$ to itself via conjugation, Then define:

$$\star = \psi \otimes (\kappa \circ *) \qquad \widetilde{\star} = \psi^{-1} \otimes (\kappa \circ *)$$

This gives:

$$\begin{aligned}\star &: W \otimes \bigwedge^p TM \otimes \overline{\bigwedge^q TM} \to W^* \otimes \bigwedge^{n-p} TM \otimes \overline{\bigwedge^{n-q} TM} \\ \widetilde{\star} &: W^* \otimes \bigwedge^r TM \otimes \overline{\bigwedge^s TM} \to W \otimes \bigwedge^{n-r} TM \otimes \overline{\bigwedge^{n-s} TM}.\end{aligned}$$

If $r = n - p$, $s = n - q$, then $\widetilde{\star} \circ \star$ is multiplication by $(-1)^{p+q}$. Now, these produce natural anti-isomorphisms of:

$$\begin{aligned}\star &: \Omega^{p,q}(W) \to \Omega^{n-p,n-q}(W^*) \\ \widetilde{\star} &: \Omega^{r,s}(W*) \to \Omega^{n-r,n-s}(W)\end{aligned}$$

Using these, we can define the "co-" of $\overline{\partial}$:

$$\delta : \Omega^{p,q}(W) \to \omega^{p,q-1}(W) \qquad \delta = -\widetilde{\star} \circ \overline{\partial} \circ \star$$

With this, if $\alpha, \beta \in A^{p,q}(W)$ then we can define:

$$(\alpha, \beta) = \langle \alpha, \star\beta \rangle = \int_M \alpha \wedge \star\beta$$

It is easy to see that $(\alpha, \alpha) \geq 0$ and $(\alpha, \alpha) = 0$ if and only if $\alpha = 0$. We also get the natural adjoint:

$$(\alpha, \delta\beta) = \langle \overline{\partial}\alpha, \beta \rangle$$

Indeed given:

$$\langle \alpha, \delta\beta \rangle = -\int_M \alpha \wedge \widetilde{\star\star}\overline{\partial}\beta = (-1)^{p+q+1}\int_M \alpha \wedge \overline{\partial} \star \beta$$

Then:

$$\begin{aligned}
(\overline{\partial}\alpha, \beta) - (\alpha, \delta\beta) &= \int_M (\overline{\partial}\alpha \wedge \star\beta + (-1)^{p+q}\alpha \wedge \overline{\partial} \star \beta) \\
&= \int_M \overline{\partial}(\alpha \wedge \star\beta) \\
&= \int_M d(\alpha \wedge \star\beta) \\
&= 0 && \text{Stokes Thm.}
\end{aligned}$$

We now reach one of the central results in Hodge theory. Define the *Hodge Laplace* (or Laplace-Bertram operator):

$$\Delta = \delta\overline{\partial} + \overline{\partial}\delta$$

This defines an operator:

$$\Delta : A^{p,q}(W) \to A^{p,q}(W)$$

Let

$$\ker \Delta = B^{p,q}(W)$$

which could be interpreted as all "complex harmonic forms", namely forms that satisfy:

$$\delta(\alpha) = \overline{\partial}(\alpha) = 0$$

The use of the elliptic operator is that it gives way to the following decomposition:

$$A^{p,q}(W) = \overline{\partial}A^{p,q-1}(W) \oplus \delta A^{p,q+1}(W) \oplus B^{p,q}(V, W)$$

Notice that we naturally get:

$$Z^{p,q}(W) = \overline{\partial}A^{p,q-1}(W) \oplus B^{p,q}(V, W)$$

Thus, when taking the Dolbeault cohomology group:

$$H^{p,q}(V, W) \cong \frac{Z^{p,q}(W)}{\overline{\partial}A^{p,q}(W)} \cong B^{p,q}(V, W)$$

Hence, the forms are naturally harmonic! This result is called the *Hodge Theorem*. Furthermore, as this is an elliptic operator, we get by elliptic operator theory that each of these spaces is *finite dimensional* ref:HERE. We thus also get:

$$H^r(M, \mathbb{C}) \cong \bigoplus_{p+q=r} B^{p,q}$$

and the Betti number is the sum of Hodge numbers:

$$b_r(V) = \sum_{p+q=r} h^{p,q}(M)$$

For future reference, an element of $H^{p+q}(M,\mathbb{Z})$ or $H^{p+q}(M,\mathbb{R})$ is said to be of type (p,q) if it is of type (p,q) when regarded as an element of $H^{p+q}(M,\mathbb{C})$.

All of this is summarized in the following two theorems:

Theorem 4.1.1: Kodaria's Finite Dimension Theorem

Then $H^{p,q}(M,E)$ is finite dimensional isomorphic to the vector space of complex harmonic forms of type (p,q) with coefficients in E. Furthermore:

$$H^p(M,E) = H^{0,p}(M,E)$$

and is zero for $p,q > n$.

Proof **:**
see a textbook in Hodge theory

Theorem 4.1.2: Serre-Hodge Duality Theorem

Let E be a complex vector bundle over the compact manifold M. Then the bilinear form $\langle -,-\rangle$ is a dual pairing of $H^{p,q}(M,E)$ with $H^{n-p,n-q}(M,E^*)$. If $K = \bigwedge^n TM$ is the canonical line bundle, Then $H^p(M,E)$ and $H^{n-q}(M,K\otimes E^*)$ are dual vector spaces.

Proof **:**
see a textbook in Hodge theory

Definition 4.1.3: Hodge Number

$$h^{p,q}(M,E) = \dim H^{p,q}(M,E)$$

If $E = \mathbb{1}$ then:

$$h^{p,q}(M) = \dim H^{p,q}(M,\mathbb{1})$$

Note that in general $h^{p,q}(M) \neq h^{q,p}(M)$; if M is a Kähler manifold this will be the case.

4.2 Defining Kähler manifold

Let M_n be a compact complex Riemann manifold. A hermitian metric on M is defined as:

$$ds^2 = 2\sum g_{\alpha\beta}(z,\overline{z})(dz^\alpha \cdot d\overline{z}^\beta)$$

where $\overline{g_{\alpha\beta}} = g_{\beta\alpha}$. To a Hermitian metric ds^2, we get a natural exterior differential form:

$$\omega = i\sum g_{\alpha\beta}(z,\overline{z})dz^\alpha \wedge d\overline{z}^\beta$$

This form can be re-written as a real differential form by taking real coordinates x^α, $1 \leq \alpha \leq 2n$ and

$$z^\alpha = x^{2\alpha-1} + ix^{2\alpha}$$

Definition 4.2.1: Kähler Metric

Let ds^2 be a Hermitian metric on the compact complex Riemann manifold M and let:

$$\omega = i \sum g_{\alpha\beta}(z, \overline{z}) dz^\alpha \wedge d\overline{z}^\beta$$

Then if $d\omega = 0$, ds^2 is called a *Kähler metric.*

If ds^2 is a Kähler metric, ω represents an element of $H^2(M, \mathbb{R})$ (using the De-Rham isomorphism); this is called the *fundamental class of the Kähler metric.*

Definition 4.2.2: Kähler Manifold

Let M be a compact complex Riemann manifold with Riemann metric ds^2. Then given a choice of ω that induces ds^2 to be Kähler metric, (M, ω) is called a *Kähler manifold.*

On a Kähler manifold, Δ commutes with the conjugation $\alpha \mapsto \overline{\alpha}$ between $B^{q,p}$ and $B^{p,q}$ which induces an isomorphism between these groups, and so:

Theorem 4.2.3: Symmetry of Hodge Number On Kähler Manifold

Let M be a Kähler manifold. Then:

$$h^{p,q}(M) = h^{q,p}(M)$$

***Proof* :**
see a textbook in Hodge Theory.

Note that these numbers do not depend on the choice of Kähler metric ($d\alpha = 0$ for a harmonic (p, q)-form regardless of metric) and hence it is enough for a Kähler metric to exist.

4.3 Euler Characteristic and Generalization

Let M be a smooth compact manifold, E a complex vector bundle (we shall introduce the Kähler assumption later). Now, as we have a double complex, we can define more than one Euler characteristic. First, recall there is the natural:

$$\chi(M, E) = \sum_{i=0}^{\infty} \dim H^i(M, E) = \sum_{i=0}^{n} \dim H^i(M, E)$$

What we can do now is chose a grading so that we can define:

$$\chi^p(M,E) = \chi\left(V, E \otimes \bigwedge^p TM\right) = \sum_{q=0}^{n} (-1)^q h^{p,q}(M,E)$$

Immediately we get:

$$\chi^0(M,E) = \chi(M,E) \qquad \text{and} \qquad \chi^p(M,E) = 0 \text{ when } p<0 \text{ or } p>n$$

and if $E = \mathbb{1}$ then:

$$\chi^p(M,\mathbb{1}) = \chi^p(M) = \sum_{q=0}^{n} (-1)^q h^{p,q}(M)$$

We can collect these characteristic in a generating function:

Definition 4.3.1: χ_y-characteristic

$$\chi_y(M,E) = \sum_{p=0}^{n} \chi^p(M,E) y^p \qquad \text{(sim.)} \qquad \chi_y(M) = \sum_{p=0}^{n} \chi^p(M) y^p$$

The term $\chi_y(M,E)$ is the χ_y-characteristic of E and $\chi_y(M)$ is called the χ_y-genus of M

By construction:

$$\chi_0(M,E) = \chi^0(M,E) = \chi(M,E) \qquad \text{(sim.)} \qquad \chi_0(M) = \chi^0(M) = \chi(M)$$

As the Euler-Hodge characteristic was called a genus, we shall call:

$$\chi(M) = \sum_{q=0}^{n} (-1)^q h^{0,q}(M)$$

the *arithmetic genus* of M. By Serre Duality:

$$\begin{aligned} \chi^p(M,E) &= (-1)^n \chi^{n-p}(M,E^*) \\ \chi(M,E) &= (-1)^n \chi(M, K \otimes E^*) \end{aligned}$$

Let us now assume M is Kähler. Let $g_q = h^{q,0}$. Then by the symmetry, the arithmetic genus $\chi(M)$ of a compact Kähler manifold is equal to:

$$\chi(M) = \sum_{i=0}^{n} (-1)^i g_i$$

Next, let us understand $\chi_y(M)$ for specific values of y. Let us start with $y=-1$. Then:

$$\chi_{-1}(M) = \sum_{p=0}^{n} (-1)^p \chi^p(M) = \sum_{p,q} (-1)^{p+q} h^{p,q}(M)$$

is equal to the Euler-Poincaré characteristic.

If V_n and V'_m are Kähler manifolds then

$$h^{p,q}(V_n \times V'_m) = \sum_{\substack{r+u=p \\ s+v=q}} h^{r,s}(V_n) h^{u,v}(V'_m).$$

Let y, z be indeterminates and associate to each Kähler manifold V the polynomial $\prod_{y,z}(V) = \sum_{p,q} h^{p,q} y^p z^q$. Then the above is equivalent to:

$$\prod_{y,z}(V_n \times V'_m) = \prod_{y,z}(V_n) \cdot \prod_{y,z}(V'_m).$$

Let $z = -1$ in. Then $\prod_{y,-1} = \chi_y$ and

$$\chi_y(V_n \times V'_m) = \chi_y(V_n) \cdot \chi_y(V'_m),$$

which is a property common to χ_y and T_y.

Now, with $y = 1$, then:

Theorem 4.3.2: Euler Characteristic With $y = 1$

$$\chi_1(M) = \sum_{p=0}^{n} \chi^p(M) = \sum_{p,q} (-1)^q h^{p,q}(M)$$

is equal to the *index* $\tau(M)$ (defined in [7, p. 8.2]).

***Proof* :**
long proof at [7, p.125, Thm 15.8.2]

This will be a key step in the proof of RR. Note that there is a more direct proof for the above that works for arbitrary compact complex manifolds M (Hirzebruch sketches a proof in [7, appendix 25.4]).

Let us take a moment to understand the first Chern class on a Kähler manifold, considering it will be key for the RR on line bundles. Take the usual exact sequence:

$$0 \to \underline{\mathbb{Z}} \to \mathcal{O}_M \to \mathcal{O}_M^* \to 0$$

where $\mathcal{O}_M$ is the sheaf of holomorphic functions, and $\mathcal{O}_M$ is the sheaf of never=zero holomorphic functions (invertible holomorphic functions). Then we get an exact cohomology sequence:

$$H^1(M, \mathcal{O}_M^*) \xrightarrow{\delta_*^1} H^2(M, \mathbb{Z}) \to H^2(M, \mathcal{O}_M^*)$$

Then on a Kähler manifold we have $H^2(M, \mathcal{O}_X^*) = H^2(M, \mathbb{1}) \cong B^{0,2}(M)$. Thus, an element $a \in H^2(M, \mathbb{Z})$ is mapped to a nonzero element of $H^2(M, \mathcal{O}_M)$ if and only if a is of type $(1,1)$. Then from characteristic theory for vector bundles we know that if $\xi \in H^1(M, \mathcal{O}_M^*)$ is a complex $\mathbb{C}^*$-bundle, then:

$$\delta_*^1(\xi) = c_1(\xi)$$

If F is a complex line bundle over M and ξ is the assocaited $\mathbb{C}^*$-bundle, then $c_1(\xi)$ is the *cohomology class* of F. Then we get:

Theorem 4.3.3: characterising Line bundles on Kähler Manifolds

Let M be a compact Kähler manifold. Then an element $a \in H^2(M, \mathbb{Z})$ is the cohomology class of a complex lien bundle over M if and only if it is of type $(1,1)$.

Proof **:**
Hirzebruch references ppl at [7, p.127, Thm 15.9.1], and says it is provable for non-Kähler manifolds and was done by Dolbeault.

Let us finish with the special case M a Kähler manifolds and $h^{p,q} = 0$ when $p \neq q$. Then $\chi_y(M)$ is (essentially) equal to the poincaré polyomial:

$$P(t; M) = \sum b_r t^r$$

where b_r is the rth Betti number of M. Indeed, we get:

$$\chi^p(M) = \sum_q (-1)^1 h^{p,q} = (-1)^p h^{p,p} = (-1) b_{2p}$$

therefore:

$$\chi_{-t^2}(M) = P(t; V) = \sum_r b_r t^r$$

Why do we care about this special case? Because complex projectives varieties $\mathbb{C}\mathrm{P}^n$ and flag manifolds $F(n)$ both are Kähler manifolds with this property. The $\mathbb{C}\mathrm{P}^n$ comes from basic Dolbeault cohomology. For the flag manifold $F(n)$, recall that $H^*(F(n), \mathbb{Z})$ by the splitting principle is generated by element $\gamma_i \in H^2(F(n); \mathbb{Z})$. Then by characteristic theory there are complex $\mathbb{C}^*$-bundles ξ_i over $Fn)$ where $c_1(\xi_i) = \gamma_i$. The "only if" comes from theorem 4.3.3 where we have that each γ_i is of type $(1,1)$, and therefore the cohomology classes of $F(n)$ are of type (p,p).

For future reference, notice that χ_y and T_y ([7, p. 14.4]) agree, as they are both essentially equal to the Poincaré polynomial.

4.3.1 Properties of χ_y-characteristic

Part of the proof will requires some splitting properties that we can induct on. To that end, we shall develop some more properties of χ_y. Let :

$$0 \to E' \to E \to E'' \to 0$$

be a short exact sequence of complex vector bundles. This naturally induces a short exact sequence of sheaves:

$$0 \to \mathcal{H}(E') \to \mathcal{H}(E) \to \mathcal{H}(E'') \to 0$$

which is left as an exercise in sheaf theory. What is important is that we get that we get that each new Euler-characteristic is additive on exact sequences:

Theorem 4.3.4: Generalized Euler Characteristic Additive

Let

$$0 \to \mathcal{H}(E') \to \mathcal{H}(E) \to \mathcal{H}(E'') \to 0$$

be a short exact sequence of complex vector bundles over M. Then:

$$\chi(M, E) = \chi(M, E') + \chi(M, E'')$$
$$\chi^p(M, E) = \chi^p(M, E') + \chi^p(M, E'')$$
$$\chi_y(M, E) = \chi_y(M, E') + \chi_y(M, E'')$$

Proof **:**
exercise.

Theorem 4.3.5: Splitting Principle and Euler Characteristic

Let E be a complex vector bundle over a compact complex manifold M, and suppose the structure group of E can be complex analytically reduced to the triangular group $\Delta(q, \mathbb{C})$. Let $A_1, A_2, ..., A_q$ be the corresponding diagonal line bundles. Let E' be another complex vector bundle over M. Then:

$$\chi(M, E' \otimes E) = \sum_i^q \chi(M, E' \otimes A_i)$$

Proof **:**
Use induction and the above theorem

Let us now get some important ways Cartier divisors come into the picture. As usual let E be a complex vector bundle over a complex manifold M, and let D be a nonsingular divisor of M. Then it is associate to a line bundle $\mathcal{O}_M(D) = \mathcal{L}(D)$. We can now tensor $E \otimes \mathcal{L}(D)$, and consider $(E \otimes \mathcal{L}(D))_D$ to be the restriction to D of the vector bundle $E \otimes \mathcal{L}(D)$. We can take the sheaf $\Omega((E \otimes \mathcal{L}(D))_D)$ over D. Extending by zero this is a sheaf on M. By the corresponding of vector bundles and sheaf, without loss of generality we may denote this sheaf by E_D.

Theorem 4.3.6: Extending via Divisors

let M be a complex manifold , D a nonsingular Cartier divisor of M, E a complex vector bundle over M. Then there is a short exact sequence:

$$0 \to \Omega(E) \to \Omega(E \otimes \mathcal{L}(D)) \to E_D \to 0$$

Proof **:**
This is a standard proof, this can be found in pure-sheaf form in Hartshorne [19, chapter 2?] or EYTNKA alg geo [1].

Let M now be compact and D still a nonsingular divisor. It now is always a compact manifold. When it is not confusing, we shall denote E_D to be the complex vector bundle over D. We shall also write:

$$\chi(D, E) := \chi(D, E_D),\ \chi^p(D, E) := \chi^p(D, E_D),\ \chi_y(D, E) := \chi_y(D, E_D)$$

Theorem 4.3.7: Splitting Euler Characteristic Using Divisor

Let M be a compact complex manifold, D a nonsingular divisor, E a complex vector bundle over M. Then:

$$\chi(M, E) = \chi(M, E \otimes \mathcal{L}(D)^{-1}) + \chi(D, E)$$

In particular, if $E = \mathbb{1}$, then:

$$\chi(M) = \chi(M, \mathcal{L}(D)^{-1}) + \chi(D)$$

Proof **:**
combine above results

Let us now apply these results to get some important identities. First, recall from sheaf theory that we get the following short exact sequence:

$$0 \to T(D) \to T(M)|_D \to \mathcal{L}(D)_D \to 0$$

We then get a natural exact sequence:

$$0 \to \bigwedge^p (T(D)) \to \bigwedge^p (T(M)|_D) \to \bigwedge^{p-1} (T(D)) \otimes \mathcal{L}(D)_D \to 0$$

Dualizing we get:

$$0 \to \bigwedge^{p-1} (T^*(D)) \otimes \mathcal{L}(D)_D^{-1} \to \bigwedge^p (T^*(M)|_D) \to \bigwedge^p (T^*(D)) \to 0$$

Now if E is a complex vector bundle over M, we can tensor each term by E_D. Then we get:

$$\chi\left(D, E \otimes \bigwedge (T^*(M))\right) = \chi^{p-1}(D, E \otimes \mathcal{L}(D)^{-1}) + \chi^p(D, E)$$

Replacing E with $E \otimes \bigwedge^p (T^*(M))$, manipulating we get:

$$\chi^p(M, E) = \chi^p(M, E \otimes \mathcal{L}(D)^{-1}) + \chi^p(D, E) + \chi^{p-1}(D, E \otimes \mathcal{L}(D)^{-1}) \tag{4.1}$$

This is true for $p > 0$, and for $p = 0$ interpret the last term on the right hand side as being 0. Furthermore, $\chi^p(D, E) = 0$ for $p = n = \dim M$ and if $p > n$ then all four terms are 0. Letting y be a indeterminate, the above equation holds with each term multiplied by y^p. Summing over $p \geq 0$ gives:

$$\chi_y(M, E) = \chi_y(M, E \otimes \mathcal{L}(D)^{-1}) + \chi_y(D, E) + y\chi_y(D, E \otimes \mathcal{L}(D)^{-1})$$

Now, by repeating the application of equation (4.1), we can write $\chi^p(D, E)$ as a linear combination of integers of the form $\chi^q(M, A)$ where A is will be an appropriate? Hirzebruch just says certain?? complex vector bundle over M. For example:

$$\chi^0(D, E) = \chi^0(M, E) - \chi^0(M, E \otimes \mathcal{L}(D)^{-1}) \tag{4.2}$$

To calculate the last term, we use Serre duality and use:

$$\chi_1(M,E) = \chi_1(M, E\otimes \mathcal{L}(D)^{-1}) + \chi_1(D,E) + \chi_1(D, E\otimes \mathcal{L}(D)^{-1})$$

and replace E with $E \otimes \mathcal{L}(D)^{-1}$ in equation (4.2) to get:

$$\chi^1(D,E) = \chi^1(M,E) - \chi^1(M, E\otimes \mathcal{L}(D)^{-1}) - \chi^0(M, E\otimes \mathcal{L}(D)^{-1}) + \chi^0(M, E\otimes \mathcal{L}(D)^{-2})$$

Recursively continuing this, we get:

$$\chi^p(D,E) = \sum_{i=0}^{p}(-1)^i[\chi^{p-i}(M, E\otimes \mathcal{L}(D)^{-i}) - \chi^{p-i}(M, E\otimes \mathcal{L}(D)^{-(i+1)})]$$

This gives an important equation. These relation's do not necessarily hold for a general complex compact manifold, but if M is Kähler, or can be embedded into projective space, the relations hold, a key fact relying on GAGA results and the isomorphism between line bundles and bundles given by divisors as shown in [1].

Combining these together in a generating function, we get:

$$\chi_y(D,E) = \sum_{i=0}^{\infty}(-y)^i[\chi_y(M, E\otimes \mathcal{L}(D)^{-i}) - \chi_y(M, E\otimes \mathcal{L}(D)^{-(i+1)})]$$

4.4 *Fundamental Kodaira Results

[7, p.138]

He now goes define a *Hodge Manifold* which is a Kähler manifolds whose fundamental class is in the image of the natural homomorphism $H^2(M,\mathbb{Z}) \to H^2(M,\mathbb{R})$. redundatnly, it was mentioned here that there are compact complex manifolds that are not Kähler. There are Kähler manifolds that are not Hodge, and but all algebraic manifold are Hodge. Note this definition is done by defining a certain order which I omitted.

Theorem 4.4.1: Bertini Theorem

Let F be a projectively induced line bundle over the algebraic manifold M. Then a non-singular divisor D induces F, namely $F = \mathcal{L}(D)$.

This is actually a special case of the general Bertini theorem that you find when googling "Bertini Theorem".

***Proof* :**
will not prove

Theorem 4.4.2: Characterizing Varieties Geoemtrically

A compact complex manifold is a smooth complete variety (Hirzebruch: algebraic manifold) if and only if it is a Hodge Manifold.

Proof **:**
not proven. It is a GAGA like result that is interesting.

Hirzebruch now states that we know that for two choices of complex vector bundles E, E' over an algebraic manifold M, it can be that $\dim H^0(M, E) \neq H^0(M, E)$. But $\chi(M, E)$ does not depend on the continuous vector bundle W and depends only on the Chern classes of E. He now introducing some vanishing theorems:

Theorem 4.4.3: Kodaira Vanishing Theorem

Let F be a complex line bundle over a compact complex manifold M. If F^{-1} is positive, then the cohomology group $H^i(M, F)$ vanish for all $i \neq n$.

I did not define positivity of a bundle, he did it at the beginning of this section

Proof **:**
proved by Kodaira

Using Serre duality, we get

Theorem 4.4.4: Kodaira Vanishing Theorem - Euler Characteristic

let F, M be as before and take $F \otimes K^{-1}$ for the canonical class K. If it is positive, then the groups $H^i(M, F)$ vanish for all $i > 0$, so that:

$$\dim h^0(M, F) = \chi(M, F)$$

Proof **:**
again Kodaira

Theorem 4.4.5: Kodaira Vanishing Theorem II

Let F be a complex line bundle over a hole manifold M, and let F' be a positive line bundle over M. Then

$$H^i(M, F \otimes (F')^k)$$

vanishg for all $i > 0$ and k sufficiently large.

Proof **:**
Kodaira.

From it we get:

Theorem 4.4.6: Characterizing Line Bundles On Algebraic Manifold

Let M be an algebraic manifold, F an analytic line bundle over M. Then there are projectively induced line bundles A, B where $F = A \otimes B^{-1}$. Thus, we have:

$$F = \mathcal{L}(A) \otimes \mathcal{L}(B)^{-1}$$

Proof **:**
Hirzbruch actually proves this, see [7, p.141, thm 18.2.5].

Hirzebruch now states that he will make no distinction between a Hodge manifold and an algebraic manifold. In particular, if a compact complex manifold M admits a Hodge metric, then M is algebraic.

Theorem 4.4.7: Line Bundles Are Algebraic Over Projective Space

let L be a $\mathbb{C}$-fiber bundle over M with $\mathbb{C}\mathrm{P}^r$ as fibers and $\mathrm{PGL}_{r+1}(\mathbb{C})$ as the structure group. Then it is an algebraic manifold.

Proof **:**
proved by Kodaira

We shall need a generization of the above for because we'll need the case where F is the flag manifold $F(q) = \mathrm{GL}_q(\mathbb{C})/\Delta(q, \mathbb{C})$ and L is the associated $\mathrm{GL}_q(\mathbb{C})$-bundle ξ over M.

Theorem 4.4.8: Borel Vanishing Theorem

Let L be a complex fibre bundle over the algebraic manifold M with an algebraic manifold as fibre and a connected structure group. Assume that the first Betti number of F is zero. Then L is itself algebraic.

Proof **:**
Borel, Hirzebruch gives a reference in [7, p. 141]

4.5 χ_y-characteristic and Splitting Principle

skipping for now

4.5.1 special case: varieties

here

5

Reduction via Cobordism

Note that if we took the GRR route, we can circumnavigate all of this with intersection theory, but it is cool for its conenction to ASIT

By Stoke's Theorem, if ω is a form on a manifold M with boundary ∂M, then:

$$\int_{\partial M} \omega = \int_M d\omega$$

Thus, figuring out the possible manifolds N such that $N = \partial M$ is a worth-while endeavor. With a bit of elementary effort, it can be shown that $N = \mathbb{R}\mathrm{P}^2$ is not the boundary of any 3-manifold M, and so the answer is false in general, but this also spurs the question of trying to classify manifolds "up to boundary". This leads to the following notion:

Definition 5.0.1: Cobordant

let M, N be two topological n manifolds. Then M, N are *cobordant* if there exists a manifold C such that

$$\partial C = M \sqcup N$$

If $M = \partial C$, then M is said to be null-cobordant (cobordant to the empty set).

The cobordism operation defines a natural ring structure (addition by cobordism, multiplication under direct product), prompting find which ring it is isomorphic to as well as find generators of the ring. This culminates in two important results:

1. The cobordism ring Ω is isomorphic to $\mathbb{Q}[x_1, x_2, ..., x_n, ...]$ where $[\mathbb{C}\mathrm{P}^i] \mapsto x_i$
2. The *signature* of Riemann manifold (see [15]) is a homomorphism from the cobordism ring to $\mathbb{Q}$

The key result this gives us is that it represents all cobordism equivalence classes using Pontryagin classes, and will be shown to reduce showing our desire result only on complex projective space.

In this section, the manifolds shall be real manifolds.

5.1 (Oriented) Cobordism Ring

Let V^n be an oriented compact differentiable manifold where n is the dimension. As V is compact and orientable, it is a cycle, and by cohomology theory $H^n(V^n, \mathbb{Z})$ is generated by one element. The value of the n-dimensional cohomology class x on the fundamental cycle V^n is denoted $x[V^n]$. If the coefficients of the cohomology group are in A, then:

$$x \in H^n(V^n, A) \qquad x[V^n] \in A$$

We can extend this by tensoring by a group B so that :

$$x[V^n] \in A \otimes B \qquad x \in H^n(V^n, A) \otimes B \cong H^n(V^n, B)$$

For our purposes, let $n = 4k$ and let $p_i \in H^{4i}(V^{4k}, \mathbb{Z})$ be the Pontryagin classes of V^{4k} (i.e. of its tangent bundle). We get a natural grading by the indices i so that given $k = j_1 + j_2 + \cdots + j_r$ we get the product $p_{j_1} \cdots p_{j_r}$ and an integer (as $A = \mathbb{Z}$):

$$p_{j_1} \cdots p_{j_r}[V^{4k}]$$

Let $\pi(k)$ be the number of distinct partitions of k.

Definition 5.1.1: Pontryagin Numbers and Chern Number

The *Pontryagin numbers* are the $\pi(k)$ numbers :

$$p_{j_1} \cdots p_{j_r}[V^{4k}]$$

where $n = 4k$. If n is not divisible by 4, then the Pontryagin number is 0.

Similarly the *Chern numbers* of a complex manifold M (with an almost complex structure so that we are given a particular orientation) are all:

$$c_{j_1} \cdots c_{j_r}[V^{2k}]$$

where $n = j_1 + j_2 + \cdots + j_r$ and the j_i run through all partitions.

By definition of the Euler class, if M_n is an oriented almost complex manifold, $c_n[M_n] = \chi(M_n, TM_n)$, showing that the Pontryagin/Chern numbers are "generalizations" of the Euler characteristic. We shall focus on Pontryagin numbers as we shall need them for the classification result, and return to Chern numbers once we cover the Todd genus in chapter 6.

We can define an equivalence relation where two manifolds $M \sim_p N$ if they have the same dimension and are the same Pontryagin number. Note that if the dimension is not divisible by 4, all manifolds are in the same equivalence relation given by $\sim_p$ as they are all 0.

If M, N are disjoint, the Pontryagin numbers of $M \sqcup N$ is the sum of the Pontryagin numbers of M, N; this comes straight from the properties of characteristic classes. Let us show the same works for the product of manifolds. Now let M, N be smooth $n = 4k$ dimensional manifolds. Then by classical differential geometry we have $T(M \times N) = \pi_M^*(TM) \times \pi_N^*(TM)$. Then:

$$(\pi_M^*(x)\pi_N^*(y))[M \times N] = x[M] \cdot y[N] \qquad \forall x \in H^n(M, \mathbb{Z}) \otimes B,\ y \in H^m(N, \mathbb{Z}) \otimes B$$

Thus, we can extend this results to characteristic sequences of Pontryagin classes. Considering the Pontryagin classes of M, N and $M \times N$:

$$\begin{gathered} 1, p_1, p_2, \ldots \\ 1, p_1', p_2', \ldots \\ 1, p_1'', p_2'', \ldots \end{gathered}$$

By the properties of characteristic classes we get:

$$1 + p_1'' + p_2'' + \cdots = (1 + p_1 + p_2 + \cdots)(1 + p_1' + p_2' + \cdots)$$

Or, as a generating function:

$$\sum_{k=0}^{\infty} p_k'' z^k = \sum_{i=0}^{\infty} \pi_M^*(p_i) z^i \sum_{j=0}^{\infty} \pi_N^*(p_j') z^j \text{ mod torsion.}$$

Thus, define the following:

Definition 5.1.2: Algebraic K-genus

Let $\{K_j\}$ be an m-sequence and let M be a 4ℓ-manifold. Then the *algebraic K-genus* of M is evaluated is:

$$K(M) = K_\ell(p_1, \ldots, p_\ell)[M] = \int_M K_\ell(p_1, \ldots, p_\ell)$$

where $p_1, \ldots, p_\ell$ are the Pontryagin classes of TM.

Proposition 5.1.3: Properties of Algebraic Genus

Let $\{K_\ell\}$ be an m-sequence, M, N be oriented 4ℓ-manifolds. Then:

1. $K(M \sqcup N) = K(M) + K(N)$
2. $K(M \times N) = K(M) \cdot K(N)$
3. $K(-M) = -K(M)$ where $-M$ is the reverse orientation put on M.

Thus, $K(-)$ respects the equivalence relation $\sim_p$.

Proof **:**
(1) and (2) follow from the above discussion, (3) follows elementary cohomology.

Definition 5.1.4: Algebraic Oriented Cobordism Ring

Let:

$$\widetilde{\Omega}^n = \{[M]_p \mid M \text{ an oriented } n\text{-manifold}\}$$

where the subscript p is dropped when it is clear. This ring is called the *algebraic cobordism ring*. By proposition 5.1.3, we can define the grade ring:

$$\widetilde{\Omega}^* = \bigoplus_{n=0}^{\infty} \widetilde{\Omega}^n$$

where:

$$\widetilde{\Omega}^n \widetilde{\Omega}^m \subseteq \widetilde{\Omega}^{nm}$$

Note that $\widetilde{\Omega}^n = 0$ if $4 \nmid n$ and $\widetilde{\Omega}^*$ is a commutative graded (torsion-free) ring.

The ring is called "algebraic" as it was defined using the algebraic properties of Pontryagin classes, we shall soon define the "geometric" cobordant ring that is the equivalence classes of cobordant manifolds and show they are isomorphic. For now, let us characterize the algebraic structure of the algebraic cobordism ring: the goal is to show that

$$\widetilde{\Omega} \otimes \mathbb{Q} \cong \mathbb{Q}[x_1, x_2, ...]$$

To that end, we must find which elements of $\widetilde{\Omega}$ can represent each x_i. There is no natural choice, hence we define "candidate sequences" of elements given by manifolds $V^4, V^8, V^{12}, ...$ that can be chosen to form this basis. We shall that the complex projective spaces of even dimension shall satisfy these required conditions. Let us start by recalling that the Pontryagin class of V^{4k} can be written in indeterminate z and decomposed into:

$$1 + p_1 z + \cdots + p_k z^k = \prod_{i=1}^{k} (1 + \beta_i z)$$

Define:

$$s(V^k) = (\beta_1^k + \cdots + \beta_k^k)[V^{4k}]$$

By some symmetry function theory, we can think of this as needing the appropriate homogeneity and dimension properties, and hence if they are nonzero for each V^{4k} they are a desired list of representatives:

Definition 5.1.5: Basis Sequence

A sequence of manifold $V^4, V^8, V^{12}, ...$ is called a *basis sequence* if:

$$s(V^{4k}) \neq 0$$

Theorem 5.1.6: Basis Sequence Correspondence

Let $\{V^{4k}\}$ be a basis sequence and B a [commutative] ring containing $\mathbb{Q}$. Then the set of sequences $\{a_k\}_k$ where $a_k \in B$ are in bijective correspondence to the set of m-sequences $\{K_j(p_1, ..., p_j)\}$ with coefficients in B where $K(V^{4k}) = a_k$.

Proof **:**
The m-sequences are in one-one correspondence (see 1.2) with power series $Q(z) = 1 + b_1 z + b_2 z^2 + \cdots$ with coefficients in B. Therefore it is sufficient to show that there is exactly one power series $Q(z)$ such that, for each V^{4k} in the sequence with PONTRJAGIN classes written in the form (5),

$$a_k = K_k[V^{4k}], \text{ where } K_k = \text{coefficient of } z^k \text{ in } \prod_{i=1}^{k} Q(\beta_i z)$$

This equation can be written

$$a_k = s(V^{4k})b_k + \text{polynomial in } b_1, b_2, \ldots, b_{k-1} \text{ of weight } k. \quad (6_k)$$

The polynomial in (6_k) depends only on V^{4k} and has integer coefficients. The coefficients b_k can now be determined uniquely by induction.

Theorem 5.1.7: Complex Projective Forms Basis Sequence

Let $\mathbb{C}\mathrm{P}^{2k}$ be the $2k$-dimensional complex projective space. Then they form a basis sequence where

$$s(\mathbb{C}\mathrm{P}^{2k+1}) = 2k + 1$$

Proof **:**
Proof: Let $h \in H^2(\mathbb{P}2k(\mathbb{C}), \mathbb{Z})$ be a generator. By 4.10.2 the PONTRJAGIN class of $\mathbb{P}2k(\mathbb{C})$ is $(1 + h^2)^{2k+1}$. The m-sequence of the power series $1 + z^k$ defines a "genus" (5.2) which for V^{4k} has the value $s(V^{4k})$ and which clearly takes the value $2k + 1$ on $\mathbb{P}_{2k}(\mathbb{C})$.

Let us now consider $\widetilde{\Omega} \otimes \mathbb{Q}$, so now $[V^{4k}]$ can be written as $\frac{a}{b}[V^{4k}]$. Certainly the Pontryagin number, K-genus, and $s(V^{4k})$ are all still well-defined (where the K-genus is well-defined as we required B to contain $\mathbb{Q}$). Now, the Pontryagin number of an element of $\widetilde{\Omega} \otimes \mathbb{Q}$ can be a non-integer. This new ring can still be represented using a basis sequence:

Theorem 5.1.8: Basis Sequence With Rational Numbers Correspondence

Let $\{V^{4k}\}$ be a basis sequence of oriented manifolds, and let $(j) = (j_1, ..., j_r)$ be a partition of k. Let $V_{(j)} = V^{4j_1} \times \cdots \times V^{4j_r}$. Then every $\alpha \in \widetilde{\Omega} \otimes \mathbb{Q}$ can be uniquely represented by the sum:

$$\alpha = \sum_{(j) \in \pi(j)} r_{(j)}[V_{(j)}] \qquad r_{(j)} \in \mathbb{Q}$$

where the sum goes over all partitions of k. Furthermore, given a system $\{a_{(j)}\}$ of rational numbers, there exists an element $\alpha \in \widetilde{\Omega} \otimes \mathbb{Q}$ where

$$p_{(j)}[\alpha] = \alpha_{(j)}$$

Proof **:**
Proof: By elementary facts on linear simultaneous equations it is sufficient to prove that a sum $\sum r_{(j)}(V_{(j)})$ over all partitions (j) of k is zero if and only if $r_{(j)} = 0$ for each (j). Suppose that $\sum r_{(j)}(V_{(j)}) = 0$. Let $q_1, q_2, q_3, \ldots$ be a sequence of indeterminates. By 6.3.1 we can find for each integer $t \geq 0$ an m-sequence which takes the value q_k^t on V^{4k}. This implies that

$$\sum_{(j)} r_{(j)} q_{(j)} = 0,$$

where $q_{(j)}$ denotes the product $q_{j_1} q_{j_2} \ldots q_{j_r}$ for $(j) = (j_1, j_2, \ldots, j_r)$. Since the $q_{(j)}$ are pairwise distinct, (8) implies that each $r_{(j)}$ vanishes (VANDERMONDE determinant).

Theorem 5.1.9: Isomorphism of Algebraic Cobordism Ring

$$\widetilde{\Omega} \otimes \mathbb{Q} \cong \mathbb{Q}[x_1, x_2, \ldots.]$$

Proof **:**
Let $\{V^{4k}\}$ be an arbitrary sequence of oriented manifolds. What's left to show is that:

1. given $\alpha = \sum_{(j)} r_{(j)}[V_{(j)}]$ then $s(\alpha) = r_k s[V^{4k}]$
2. If for all k, every element of $\alpha \in \widetilde{\Omega}^{4k} \otimes \mathbb{Q}$ can be represented as a sum $\alpha = \sum_{(j)}[V_{(j)}]$, then $\{V^{4k}\}$ is a basis sequence

Proof of I): Let $\{K_j\}$ be the m-sequence of the power series $1 + z^k$. This m-sequence takes the value $s(\alpha)$ on elements $\alpha \in \Omega^{4k} \otimes Q$ and the value 0 on elements of $\Omega^{4j} \otimes Q$ with $1 \leq j < k$. This implies (7*).

Proof of II): Suppose that, for some k, $s(V^{4k}) = 0$. Then, by I), $s(\alpha) = 0$ for all $\alpha \in \Omega^{4k} \otimes Q$. But $s(\mathbf{P}_{2k}(\mathbf{C})) = 2k + 1$ by 6.3.2. Contradiction.

There is a natural question of smallest integers can represent a system of Pontryagin numbers and it invovles the μ that I skipped, but it is not important for our purposes (see [7, section 6.4] for details).

Lastly, the K-genus can be characterized as as representing homomorphism from the algebraic cobordism ring to $\mathbb{Q}$:

Theorem 5.1.10: Characterizing Rational Homomorphisms

Let $\varphi : \widetilde{\Omega} \otimes \mathbb{Q} \to \mathbb{Q}$ be a homomorphism. Then there exists a K such that $\varphi(-) = K(-)$.

We shall use this property to define the geometric K-genus.

Proof **:**
easy on p.81

This fully encodes the algebraic information. Let us now define the geometric counter-part

Definition 5.1.11: Geometric Oriented Cobordism Ring

Two oriented smooth n-dimensional manifolds M^n, N^n are *cobordant* if there exist a manifold C such that

$$\partial C \cong (M^n \sqcup -N^n)$$

where $-N^n$ is the reverse orientation; if such a C exists it defines an equivalence relation $M \sim_c N$ which respects disjoint union, reverse orientation, and direct product; $\sqcup, -, \times$. Then Ω^n is the collection of such equivalence classes and:

$$\Omega = \bigoplus_n \Omega^n$$

is the *geometric cobordism ring.*

Definition 5.1.12: Geometric Genus

Let Ω^* be the oriented geometric cobordism ring. Then a *Geometric genus* (with respect to B) is a B-homomorphism:

$$\varphi : \Omega^* \to B$$

After showing Ω^* and $\widetilde{\Omega}^*$ are isomorphic, we shall see that all geometric genii are representable by an m-sequence .

Certainly this ring is graded, and it is *anti-commutative*, namely:

$$\alpha \cdot \beta = (-1)\beta \cdot \alpha \qquad \alpha \in \Omega^n,\ \beta \in \Omega^m$$

He now goes onto to prove the isomorphism between this ring and the algebraic one, skipping for now.

5.2 Intersection Form: Index and Hirzebruch Signature Theorem

On any $4k$-dimensional manifold, there is a natural generalization of the symplectic form known as the *intersection form*: given a choice of $x, y \in H^{2k}(M, \mathbb{R})$, Define:

$$Q(x, y) = x \smile y[M^n]$$

As $x \smile y \in H^4k$:

$$(x \smile y) = (-1)^{k^2}(y \smile x) = (y \smile x)$$

show that the form is *symmetric* (rather than anti-symmetric), which is why we are working with 4-dimensional manifolds. Thus, the eigenvalues of the corresponding matrix representations are always *real* and there always exists an orthogonal basis. If e_p and e_n represents the number of positive and negative eigenvalues, then:

Definition 5.2.1: Index

$$\tau(Q) = e_p - e_n$$

The most important result about the index is that it is a *genus*:

Theorem 5.2.2: Index is Genus

Let τ be the index function. Then:

1. $\tau(M \sqcup N) = \tau(M) + \tau(N)$
2. $\tau(M) \times N) = \tau(M) \cdot \tau(N)$
3. $\tau(-M) = -\tau(M)$
4. If M is nullcobordant, $\tau(M) = 0$

Proof **:**
Proof. I) follows immediately from the definitions of $V^n + W^n$ and $-V^n$.

II) is known (THOM [2]) but is given there without proof. We therefore prove II) in full. Let $M^{4k} = V^n \times W^m$. Then

$$H^{2k}(M^{4k}, \mathbf{R}) \cong \sum_{s=0}^{2k} H^s(V^n, \mathbf{R}) \otimes H^{2k-s}(W^m, \mathbf{R}).$$

Elements $x, y \in H^{2k}(M^{4k}, \mathbf{R})$ are said to be orthogonal if $xy[M^{4k}] = 0$. Introduce bases $\{v_i\}$ for $H^s(V^n, \mathbf{R})$ and $\{w_j\}$ for $H^t(W^m, \mathbf{R})$ such that $v_i^s v_j^{n-s}[V^n] = \delta_{ij}$ for $s \neq \frac{n}{2}$ and $w_i^t w_j^{m-t}[W^m] = \delta_{ij}$ for $t \neq \frac{m}{2}$.

Now consider the group $A = H^{\frac{n}{2}}(V^n, \mathbf{R}) \otimes H^{\frac{m}{2}}(W^m, \mathbf{R})$, taking $A = 0$ if n and m are odd. Then A is orthogonal to the subgroup B of $H^{2k}(M^{4k}, \mathbf{R})$, which consists of all elements of the summation (1) in which no elements of A occur. As a basis for the group B we can take $\{v_i^s \otimes w_j^{2k-s}\}, (0 \leq s \leq n, s \neq \frac{n}{2})$. Now

$$\begin{aligned}(v_i^s \otimes w_j^{2k-s})(v_{i'}^{s'} \otimes w_{j'}^{2k-s'})[M^{4k}] &= \pm 1 \text{ if } s + s' = n, i = i', j = j' \\ &= 0 \text{ otherwise.}\end{aligned}$$

It follows that, with respect to this basis, the restriction of the bilinear form $xy[M^{4k}]$ to B is represented by a matrix with blocks $\pm\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ down the diagonal and zero elsewhere. Therefore the index of the restriction of $xy[M^{4k}]$ to B is 0. Since A and B are orthogonal, $\tau(M^{4k})$ is equal to the index $\tau(A)$ of the restriction of the bilinear form $xy[M^{4k}]$ to A. There are now two cases to consider. If n and m are not divisible by 4 then $\tau(A) = 0$. If n and m are divisible by 4 then $\tau(A) = \tau(V^n) \cdot \tau(W^m)$. This completes the proof of II). A more detailed proof can be found in CHERN-HIRZEBRUCH-SERRE [1].

III) is proved by THOM [1]. The proof can be summarised briefly as follows. Suppose that V^{4k} is the oriented boundary of X^{4k+1}, and that $j : V^{4k} \to X^{4k+1}$ is the embedding. THOM considers the diagram of homomorphisms

$$\begin{tikzcd} H^{2k}(X^{4k+1},\mathbf{R}) \arrow[r, "j^*"] \arrow[d] & H^{2k}(V^{4k},\mathbf{R}) \arrow[r] \arrow[d, "i"] & H^{2k+1}(X^{4k+1} \bmod V^{4k},\mathbf{R}) \arrow[d] \\ H_{2k+1}(X^{4k+1} \bmod V^{4k},\mathbf{R}) \arrow[r] & H_{2k}(V^{4k},\mathbf{R}) \arrow[r, "j_*"] & H_{2k}(X^{4k+1},\mathbf{R}) \end{tikzcd}$$

Here the rows are part of exact homology and cohomology sequences, and the vertical arrows are isomorphisms, defined by Poincaré duality for V^{4k} and X^{4k+1}, which make the squares commutative.

Let A^{2k} be the image of j^* in $H^{2k}(V^{4k},\mathbb{R})$ and let K_{2k} be the kernel of j_* in $H_{2k}(V^{4k},\mathbb{R})$. Then A^{2k} is dual to $H_{2k}(V^{4k},\mathbb{R})/K_{2k}$ under the duality between $H^{2k}(V^{4k},\mathbb{R})$ and $H_{2k}(V^{4k},\mathbb{R})$. On the other hand, the diagram implies that, for $x \in H^{2k}(V^{4k},\mathbb{R})$,

$$x \in A^{2k} \Leftrightarrow i(x) \in K_{2k}.$$

Therefore, if $b_{2k} = \dim H_{2k}(V^{4k},\mathbb{R})$ is the $2k$-th Betti number of V,

$$\dim A^{2k} = \dim K_{2k} = b_{2k} - \dim K_{2k}$$

and

$$\dim A^{2k} = \frac{1}{2}b_{2k}.$$

If $x = j^*y \in A^{2k}$, and v is the fundamental cycle of V^{4k} then $x^2[V^{4k}] = (j^*y^2)[v] = (y^2)[j_*v] = 0$. Therefore the cone $\{x \in H^{2k}(V^{4k},\mathbb{R}); x^2[V^{4k}] = 0\}$ contains the linear subspace A^{2k} of dimension $\frac{1}{2}b_{2k}$. It follows that the bilinear form $xy[V^{4k}]$ has $p^+ = p^-$ (8.1) and hence that $\tau(V^{4k}) = 0$. This proves III) and completes the proof of Theorem 8.2.1.

Thus, by theorem 5.1.10, there is an m-sequence that represents this map, immediately giving:

Theorem 5.2.3: Hirzebruch Signature Theorem

Let M be a real differentiable oriented manifold, and let $I(M)$ be its index. Then:

1. if $\dim_{\mathbb{R}} M \not\equiv 0 \bmod 4$, then
$$I(M) = 0$$
2. if $\dim_{\mathbb{R}} M \equiv 0 \bmod 4$, then:
$$I(M) = L_k(p_1, ..., p_k)[M]$$

***Proof* :**
We must show they match on the generators $\mathbb{C}\mathrm{P}^n$. To compute the index on $\mathbb{C}\mathrm{P}^n$. [10, p.231]. Proof. Recall, I is a function from manifolds to $\mathbb{Z}$ and clearly satisfies (1). By Thom's second Theorem (Theorem 3.52), I satisfies (2). Take as basis sequence: $M_{4k} = \mathbb{P}^{2k}_{\mathbb{C}}$. We have

$$I(M_{4k}) = \sum_{p=0}^{2k} (-1)^p h^{p,q}(M_{4k}),$$

by the Hodge Index Theorem (Theorem 2.77). As $h^{p,p} = 1$ and $h^{p,q} = 0$ if $p \neq q$, we get

$$I(M_{4k}) = 1.$$

Now we further know the Künneth formula for the $h^{p,q}$ of a product (of two, hence any finite number of complex manifolds). Apply this and get (DX)

$$I(\mathbb{P}_{\mathbb{C}}^{j_1} \prod \cdots \prod \mathbb{P}_{\mathbb{C}}^{j_r}) = 1.$$

Therefore, (3) holds. Then, our previous theorem implies $I(M) = K(M)$ for some K, a multiplicative sequence. But, $K(\mathbb{P}_{\mathbb{C}}^{2k}) = 1$, there and we know there is one and only one multiplicative sequence $\equiv 1$ on all $\mathbb{P}_{\mathbb{C}}^{2k}$, it is L. Therefore, $I(M) = L$, as claimed.

5.3 Derived Index

This is about the index of the intersection of manifolds that remember the manifolds that where intersected. This can be done within the modern language of *derived algebraic geometry* by taking $M \otimes^{\mathbf{L}} N$, but there is no point in building up an entire framework and never use its results, just state the results within its language. Instead, we shall

Let M^n be a compact oriented smooth manifold, $j : V^{n-k} \to M^n$ an embedding of the oriented submanifold V^{n-k}. Let $N_{M/V}$ be the normal bundle of V with respect to M. if j^* is the cohomology pullback

Then by some basic Riemannian geometry we have that we can embed TV into TM and we have:

$$TM = TV \oplus NV$$

where $NV := N_{M/V}$ is the normal bundle. Thus, of p is the total Pontryagin class:

$$j^* p(TM) = p(TV) \cdot p(NV) \text{ mod torsion}$$

By **??**:

$$p(NV) = j^*(1 + v^2)$$

Thus:

$$p(V) = j^* \left((1 + v^2)^{-1} p(M)\right)$$

Now, taking the L-genus and taking advantage of the multiplicative and naturality property we get:

$$L(j^*(TM) = j^*(L(TM)) = L(TV \oplus NV) = L(TV) \cdot T(NV)$$

Thus we get:

$$j^*(L(TM)) = L(TV)) \cdot L(NV)$$

Moving things around:

$$L(TV) = j^*(L(TM)) \cdot L(NV)^*$$

Since we know the normal bundle has a single Pontryagin root $\gamma = v^2$, we get:

$$L(NV) = \frac{v}{\tanh(v)}$$

Finally, as $L(TM) = \sum_i^\infty L_i(TM) = \sum_i^\infty L_i(p_1(TM), ..., p_i(TM))$, we get:

$$L(TV) = \sum_{i=0}^{\infty} L_i(p_1(V), ..., p_i(V)) = j^* \left[\frac{\tanh(v)}{v} \sum_{i=0}^{\infty} L_i(p_1(M), ...p_i(M) \right]$$

Using this, we take advantage of Hirzebruch's Signature Theorem to get $\tau(V)$. First, by Poincaré duality, if $x \in H^{n-2}(M, \mathbb{Z}) \otimes \mathbb{Q}$, then:

$$j^*(x)[V] = vx[M]$$

Now, by Hirzebruch's signature theorem:

$$\tau(V) = \int_M \tanh \sum_{i=0}^{\infty} L_i(p_1(M), ..., p_i(M)$$

Using the computational methods available to us we get:

$$\begin{aligned}
n = 2 \qquad & \tau(V^0) = \int_{M^2} v \\
n = 6 \qquad & \tau(V^4) \int_{M^6} \frac{1}{3}(-v^3 + p_1 v) \\
n = 10 \qquad & \tau(V^8) = \int_{M^{10}} \frac{1}{45}(6v^5 - 5p_1 v^3 + (7p_2 - p_1^2)v)
\end{aligned}$$

Now here is where interpreting Chern classes naturally in intersection theorem comes in. Say we have a collection $v_1, ..., v_r \in H^2(M, \mathbb{Z})$. v_1 is associated to V^2. Now say v_2 is restricted to V^{n-2} and is associated to $V^{n-4} \subseteq V^{n-2}$. Finally, v_r is restricted to $V^{n-2(r-1)}$ and is associated to V^{n-2r}. Then if $j : V^{n-2r} \to M^n$, then:

$$j^*(x)[V^{n-2r}] = v_1 v_2 \cdots v_r x[M^n]$$

Then applying the L-power series successfully, we get the index formula:

$$\tau(V^{n_2 r}) = \int_{M^n} \tanh v_1 \tanh v_2 \cdots \tanh v_r \sum_{i=0}^{\infty} L_i(p_1(M^n), ..., p_i(M^n)$$

Then for sure by [5], it was proven that every 2-dimensional integral cohomology clsas of a compact oriented differentiable manifold M^n can be represented by a submanifold $V^{n-2} \subseteq M^n$. By applying this result iteratively, we can justify the above equation. Thus, $\tau(V^{n-2r})$ depends only on the (unordered) choice of cohomology classes $v_1, v_2, ..., v_r$. Thus, we can write:

Definition 5.3.1: Virtual Index

$$\tau(v_1, ..., v_r) = \int_{M^n} \tanh v_1 \tanh v_2 \cdots \tanh v_r \sum_{i=0}^{\infty} L_i(p_1(M^n), ..., p_i(M^n)$$

Every virtual index occurs as the index of a submanifold of M^n and is thus an integer.

Recall tanh satisfies:

$$\tanh(u+v) = \frac{\tanh(u)+\tanh(v)}{1+\tanh(u)\tanh(v)}$$

or as Hirzebruch writes it:

$$\tanh(u+v) = \tanh(u) + \tanh(v) - \tanh(u)\tanh(v)\tanh(u+v)$$

This immediately gives: Here's the LaTeX code for the theorem:

Theorem 5.3.2: Functional Equation of Virtual Index

The virtual index is a function which associates an integer $\tau(v_1, v_2, \ldots, v_r)$ to each (unordered) r-ple $(v_1, v_2, \ldots, v_r)$ of 2-dimensional integral cohomology classes of a compact oriented differentiable manifold M^n. The function τ is zero if $n - 2r \not\equiv 0$ modulo 4, if $2r > n$, or if one of the classes v_i is zero. It satisfies the functional equation

$$\tau(v_1, \ldots, v_r, u+v) = \tau(v_1, \ldots, v_r, u) + \tau(v_1, \ldots, v_r, v) - \tau(v_1, \ldots, v_r, u, v, u+v)$$

If $n = 4k+2$:

$$\tau(u+v) = \tau(u) + \tau(v) - \tau(u, v, u+v)$$

(section 9.4 seems not important)

6

Todd Class

In section 3.4, we defined the Chern character which gives:

$$\mathrm{ch}(E_1 \oplus E_2) = \mathrm{ch}(E_1) + \mathrm{ch}(E_2) \in H^*(X, \mathbb{Q})$$

In section 4.3, we saw that we can define the Euler characteristic for vector bundle:

$$\chi(X, E_1 \oplus E_2) = \chi(X, E_1) + \chi(X, E_2)$$

This algebraic property gives us the hope that we can directly generalize Chern's Theorem to:

$$\chi(X, E) = \int_X \mathrm{ch}(E)$$

However, this does not give the expected result:

Example 6.0.1: Incorrect Generalization
Let $E = \mathbb{1}$ be the trivial bundle. Then $\mathrm{ch}(\mathbb{1}) = 1$. Now consider $X = S^2$. Then:

$$2 = \chi(S^2) = \chi(S^2, \mathbb{1}) \neq \int_{S^2} \mathrm{ch}(S^2) = 0$$

showing we do not recover the Euler characteristic.

The problem is that we are not taking into considering the geometric information of X. What would be desired is a "correction class" $C(TX)$ (C for correction) where TX is the tangent bundle where:

$$\chi(X, E) = \int_X \mathrm{ch}(E) \smile C(TX)$$

This shall be the *Todd class* $\mathrm{td}(X)$ where $\mathrm{td}(X) = C(TX)$. What properties should td require? For one, we must have:

$$\chi(X, \mathbb{1}) = \chi(X) = \int_X \mathrm{td}(X)$$

For another, as $\chi(X \oplus Y) = \chi(X) \cdot \chi(Y)$, $\mathrm{td}(X \oplus Y) = \mathrm{td}(X) \cdot \mathrm{td}(Y)$, namely it is *multiplicative* on direct sums. Finally, as all compact complex manifolds are "built" from the complex projective spaces as demonstrated in the cobordism isomorphism in theorem 5.1.9, it is necessary to have $\mathrm{td}(\mathbb{C}\mathrm{P}^k)$ agree with $\chi(\mathbb{C}\mathrm{P}^k)$. In particular, as we know the line bundles over $\mathbb{C}\mathrm{P}^k$, we can deduce td by making it agree with the values of χ over each characteristic.

To that end, let us show what it has to be the case where $X = \mathbb{C}\mathrm{P}^1$ so that $T\mathbb{C}\mathrm{P}^1 = \mathcal{O}(2)$, the twisted line bundle, and $\chi(\mathbb{C}\mathrm{P}^1) = 1$. To define $\mathrm{td}(\mathbb{C}\mathrm{P}^1)$ (which again is inputting the tangent bundle $T\mathbb{C}\mathrm{P}^1$), we shall need to define a power series (the details of why are in section 6.1), in particular we shall have:

$$Q(x) = 1 + a_1 x + a_2 \frac{x^2}{2!} + a_3 \frac{x^3}{3!} + \cdots$$

and have $\mathrm{td}(\mathbb{C}\mathrm{P}^1) = Q\left(c_1(\mathcal{O}(2))\right)$. It is easy to compute that

$$c_1(\mathcal{O}(2)) = h \in H^2(\mathbb{C}\mathrm{P}^1; \mathbb{Z}) \cong \mathbb{Z}[h]/(h^2)$$

as $h^2 = 0$, we get:

$$\begin{aligned}
\chi(\mathbb{C}\mathrm{P}^1) = 1 &= \int_{\mathbb{C}\mathrm{P}^1} \mathrm{td}(\mathbb{C}\mathrm{P}^1)) \\
&= \int_{\mathbb{C}\mathrm{P}^1} Q(c_1(T\mathbb{C}\mathrm{P}^1)) \\
&= \int_{\mathbb{C}\mathrm{P}^1} Q(2h) \\
&= \int_{\mathbb{C}\mathrm{P}^1} (1 + 2a_1 h) = 2a_1
\end{aligned}$$

Thus, it must be that $a_1 = \frac{1}{2}$, giving us:

$$Q(x) = 1 + \frac{1}{2}x + a_1 \frac{x^2}{2!} + \cdots$$

Notice that if we consider $\mathcal{O}(\mathbb{C}\mathrm{P}^1, \mathcal{O}(d))$, then we get:

$$\chi(\mathbb{C}\mathrm{P}^1, \mathcal{O}(d)) = d + 1$$

and to compute

$$\mathrm{ch}(\mathcal{O}(d)) = e^{kh} = 1 + dh$$

as $h^2 = 0$. Thus:

$$\begin{aligned}\chi(\mathbb{C}\mathrm{P}^1, \mathcal{O}(d) = d+1 &= \int_{\mathbb{C}\mathrm{P}^1} \mathrm{ch}(\mathcal{O}(d)) \smile \mathrm{td}(\mathbb{C}\mathrm{P}^1)) \\ &= \int_{\mathbb{C}\mathrm{P}^1} \mathrm{ch}(\mathcal{O}(d)) \smile Q(c_1(T\mathbb{C}\mathrm{P}^1)) \\ &= \int_{\mathbb{C}\mathrm{P}^1} \mathrm{ch}(\mathcal{O}(d)) \smile Q(2h) \\ &= \int_{\mathbb{C}\mathrm{P}^1} (1+2h) \smile (1+h) \\ &= \int_{\mathbb{C}\mathrm{P}^1} [1+(1+d)h] = d+1\end{aligned}$$

Giving equality in this special case. Repeating this process again for $\mathbb{C}\mathrm{P}^k$ for $k \in \mathbb{N}$, we shall get the next coefficients. In section 6.1, we go over the general algebraic method for finding such a power series, and in proposition 6.1.4 we shall show how we deduce the rest of this Todd power series. Overall, repeating this process will define td and we by construction have it agree on at least the Euler characteristic of $\mathbb{C}\mathrm{P}^n$ with the standard line bundles. We shall show it is enough in the proof of Hirzebruch Riemann Roch by taking advantage of the results on Cobordism and the splitting principle.

After defining the Todd class, we shall combine the Todd class and Chern character to create the *T-characteristic* and again show how it well-defined and has the desired properties for any vector bundle over a space $E \to X$.

6.1 Algebraic Build-up: Multiplicative sequences

As we saw with Chern and Pontryagin classes, the direct sum of bundles translates to a cup product of characteristic classes. This gives a natural algebraic relation that we know codify for future reference. Let B be a commutative ring, and define:

$$\mathcal{B} = B[z, p_1, p_2, ...]$$

Define a grading given by the following:

$$p_{j_1} p_{j_2} \cdots p_{j_r} \text{ has weight } j_1 + j_2 + \cdots + j_r$$

Then let $\mathcal{B}_k$ be the weight k-groups; note $\mathcal{B}_0 = B$ and $\mathcal{B}_r \mathcal{B}_s = \mathcal{B}_{r+s}$. Let $\pi(k)$ be all partitions of k. Then $\mathcal{B}_k$ is a B-module of rank $\pi(k)$.

Definition 6.1.1: Multiplicative Sequence

Let $\{K_j\}$ be a sequence of polynomials in indeterminates p_i where $K_0 = 1$ and $K_j \in \mathcal{B}_j$. Then the sequence $\{K_j\}$ is called a *multiplicative sequence* or *m-sequence* if whenever a power series is the product of two power series:

$$1 + p_1 z + p_2 z^2 + \cdots = (1 + p_1' z + p_2' z^2 + \cdots)(1 + p_1'' z + p_2'' z^2 + \cdots)$$

with indeterminates z, p_i', p_i'', then we get the identify:

$$\sum_{j=0}^{\infty} K_j(p_1, p_2, ...p_j) z^j = \left(\sum_{i=1}^{\infty} K_i(p_1', p_2', ...p_i') z^i \right) \left(\sum_{j=1}^{\infty} K_j(p_1'', p_2'', ...p_i'') z^j \right)$$

that is:

$$K\left(\sum_{i=0}^{\infty} q_i z^i \right) = \sum_{i=0}^{\infty} K_i(q_1, ..., q_i) z^i$$

where $q_i \in \mathcal{B}$ of weight i. In other words a sequence $\{K_j\}$ is called multiplicative if it induces an endomorphism on

$$B[p_1, p_2, ...]\,[[z]]$$

Given a multiplicative sequence $\{K_j\}$, we associate to it its *characteristic power series* as the image of $K(1+z)$:

$$K(1+z) = \sum_{j}^{\infty} K_j(1, 0, ..., 0) z^j = \sum_{j}^{\infty} b_j z^j$$

An important property of vector bundles is that they can be pulled back over a flag manifold so that they split into line bundles. This strategy shall be important to us, and hence we will be considering:

$$1 + p_1 z + \cdots + p_n z^n = \prod_{i}^{n} (1 + \beta_i z)$$

where β_i is a symmetric polynomial in the p_i's. By some elementary ring theory and theory of symmetric functions, we can interpret $\mathcal{B}$ as the ring of all symmetric functions in $\beta_1, ..., \beta_n$.

We can characterize m-sequences via 1-unit power series:

Proposition 6.1.2: Correspondence 1-unit Power Series and m-sequences

There is a bijective correspondence between m-sequences and formal power series with constant term 1.

***Proof* :**

First, let's show any m-sequence $\{K_j\}$ is completely determined by its characteristic power series

$$Q(z) = K(1+z)$$

Proof. By decomposing:

$$1 + p_1 z + \cdots + p_n z^n = \prod_i^n (1 + \beta_i z)$$

We get:

$$\sum_{j=0}^{m} K_j(p_1, \ldots, p_j) z^j + \sum_{j=m+1}^{\infty} K_j(p_1, \ldots, p_m, 0, \ldots, 0) z^j = \prod_{i=1}^{m} Q(\beta_i z). \quad (6_m)$$

Therefore any polynomial K_j with $j \leq m$ is determined as a symmetric polynomial in the β_i and hence as a polynomial in the p_i. This holds for arbitrary m, completing the proof. $\square$

Next, let

$$Q(z) = 1 + \sum_{i=1}^{\infty} b_i z^i \qquad b_i \in B$$

be a formal power series. Then there is an associated m-sequence $\{K_j\}$ where $K(1+z) = Q(z)$.

Proof. Consider the usual decomposition:

$$1 + p_1 z + \cdots + p_n z^n = \prod_i^n (1 + \beta_i z)$$

and consider the product $\prod_{i=1}^{m} Q(\beta_i z)$ where β_i are symmetric functions. Then the coefficients z^j are homogeneous of weight j. Thus, it is in $K_j^{(m)}(p_1, \ldots, p_j)$. It follows that $K_j^{(m)}$ does not depend on m for $m \geq j$.

Now, define $K_j = K_j^{(m)}$ for $m \geq j$. I claim $\{K_j\}$ is the required m-sequence. Indeed, the earlier proof the multiplicative property is true if the indeterminates p'_i, p''_i are replaced by 0 for large values of i, and so $\{K_j\}$ is an m-sequence. Finally applying $m = 1$ in equation:

$$\sum_{j=0}^{m} K_j(p_1, \ldots, p_j) z^j + \sum_{j=m+1}^{\infty} K_j(p_1, \ldots, p_m, 0, \ldots, 0) z^j = \prod_{i=1}^{m} Q(\beta_i z). \quad (6_m)$$

shows that $K(1+z) = Q(z)$, completing the proof. $\square$

Combining these, we get the correspondence

The simplest example would be $\{p_j\}$ which has the characteristic series $1+z$. The notation p_i is due to the connection to Pontryagin classes. Recall Pontryagin classes are given in the $4i$th cohomology group. The HRR theorem shall work with Chern classes which are in the $2i$th cohomology gorups, and so we shall be working with the following will be a useful substitution. Let $c_0 = p_0 = 1$, $z = x^2$, and γ_i be such that $\beta_i = \gamma_i^2$. then:

$$\sum_{i=1}^{\infty} p_i (-z)^i \left(\sum_{j=0}^{\infty} c_j (-x)^j \right) \left(\sum_{i=0}^{\infty} c_i x^i \right)$$

Naturally, if $\{K_j(p_1, ..., p_j)\}$ is an m-sequence with $Q(z)$ as the characteristic power series, then let

$\{\widetilde{K}_j(c_1, ...c_j)\}$ be an m-sequence where $\widetilde{Q}(x) = Q(x^2)$. Then:

$$K_j(p_1, ..., p_j) = \widetilde{K}_{2j}(c_1, ..., c_{2j})$$
and
$$0 = \widetilde{K}_{2j+1}(c_1, ..., c_{2j+1})$$

Thus, the m-sequence in c_i with the characteristic power series $1 + x^2$ is given by

$$1, 0, p_1, 0, p_2, ...$$

From this relations, we get:

$$\begin{aligned}
p_1 &= -2c_2 + c_1^2 \\
p_2 &= 2c_4 - 2c_3c_1 + c_2^2 \\
p_3 &= -2c_6 + 2c + 5c + 1 = 2c_4c_2 + c_3^2, ...
\end{aligned}$$

Let's now find a way to calculate the explicit calculation of the polynomials of m-sequences. Given $Q(z) = 1 + \sum_{i=1}^{\infty} b_i z^i$, consider the factorization:

$$1 + b_1 z + b_2 z^2 + \cdots + b_m z^m = (1 + \beta_1' z) \cdots (1 + \beta_m' z)$$

Then define the sum:

$$\sum (\beta_1')^{j_1} (\beta_2')^{j_2} \cdots (\beta_r')^{j_r} \qquad \text{where} \qquad j_1 \geq j_2 \geq \cdots \geq j_r \geq 1, \ \sum_{s=1}^{r} j_s = k \leq m$$

which denotes the symmetric function in the β_i' which is the sum of all pairwise distinct monomials obtained by permuting $(\beta_1', ..., \beta_m')$ to the monomial $(\beta_1')^{j_1} (\beta_2')^{j_2} (\beta_r')^{j_r}$. The number of monomial in the sum is $m!/h$, h being the number of permutations living the aforementioned monomial fixed. Now:

Lemma 6.1.3: Computation of m-sequence

Let $\{K_j(p_1...., p_j)\}$ be the m-sequence corresponding to $Q(z) = 1 + \sum_{i=1}^{\infty} b_i z^i$. Then the coefficient of $p_{j_1} p_{j_2} \cdots p_{j_r}$ of K_s where $j_1 \geq j_2 \geq \cdots \geq j_r \geq 1$ and $\sum_{s=1}^{r} j_s = k$ is equal to:

$$\sum (j_1, j_2, ..., j_r)$$

***Proof* :**
it is computations that will be omitted.

Define:

$$Q(z) = \frac{\sqrt{z}}{\tanh \sqrt{z}} = 1 + \sum_{k=1}^{\infty} (-1)^{k-1} \frac{2^{2k}}{(2k)!} B_k z^k$$

where the B_k are the Bernoulli numbers. Note that in this case the coefficient ring is $B = \mathbb{Q}$. Furthermore, we shall have $B_k > 0$ skip the value of $1/2$. Thus we get:

$$B_1 = \frac{1}{6}, \ B_2 = \frac{1}{30}, \ B_3 = \frac{1}{42}, \ B - 4 = \frac{1}{30}, \ B_5 = \frac{5}{66}, \ B_6 = \frac{691}{2730}, \ B_7 = \frac{7}{6}, \ B_8 = \frac{3617}{510}, ...$$

We shall denote the m-sequence of $Q(z)$ as $\{L_j(p_1, ..., p_j)\}$. Computing we get:

$$\begin{aligned} L_1 &= \frac{1}{3}p_1 \\ L_2 &= \frac{1}{5\cdot 7}(7p_2 - p_1^2) \\ L_3 &= \frac{1}{3^3\cdot 5\cdot 7}(62p_3 - 13p_2p_1 + 2p_1^3) \\ &\vdots \end{aligned}$$

We will want the following normalization: $L_k(p_1...., p_k) = 1$. To that end, if we let $p_i = \binom{2k+1}{i}$, then we get:

$$1 + p_1z + p_2z^2 + \cdots + p_kz^k = (1+z)^{2k+1} \bmod z^k + 1$$

From this we get:

Proposition 6.1.4: Coefficient of L-characteristic Series

Let $Q(z) = \frac{\sqrt{z}}{\tanh\sqrt{z}}$. Then for every k, the coefficient J_k of z^k in $(Q(z))^{2k+1}$ is equal to 1, and $Q(z)$ is the *only* power series with rational coefficient with this property.

***Proof* :**
Apply Cauchy's integral formula and compute (see [7, p.12] to see the exact computations)

The above is useful for Pontryagin classes. Let us find the equivalent for Chern classes. Let:

$$Q(x) = \frac{x}{1-e^{-x}} = 1 + \frac{1}{2}x + \sum_{k=1}^{\infty}(-1)^{k-1}\frac{B_k}{(2k)!}x^{2k}$$

be the characteristic polynomial. Then we get an m-sequence of polynomials $\{T_k\}$ called *Todd polynomials.* For future reference, we have the identity:

$$\frac{x}{1-\exp(-x)} = \exp\left(\frac{1}{2}x\right)\frac{\frac{1}{2}x}{\sinh\frac{1}{2}x}$$

The m-sequence for $Q(x) = \frac{\frac{1}{2}x}{\sinh\frac{1}{2}x}$ is given by:

$$A_1 = \frac{-2}{3},\ A_2 = \frac{2}{45}(-4p_2 + 7p_1^2),\ A_3 = \frac{-4}{3^3\cdot 5\cdot 7}(16p_3 - 44p_2p_1 + 31p - 1^3),\ \cdots$$

By using the translation from Pontryagin to Chern class lemma, we get:

$$T + k(c_1, ..., c_k) = \sum_{\substack{r,s \\ r+2s=k}} \frac{1}{2^{4s}r!}\left(\frac{1}{2}c_1\right)^r A_s(p_1, ..., p_s)$$

We can compute the first few values:

$$\begin{aligned}
T_1 &= \frac{1}{2}c_1 \\
T_2 &= \frac{1}{12}(c_2 + c_1^2) \\
T_3 &= \frac{1}{24}c_2c_1 \\
T_4 &= \frac{1}{720}(-c_4 + c_3c_1 + 3c_2^2 + 4c_2c_1^2 - c_1^4) \\
T_5 &= \frac{1}{1440}\left(-c_4c_1 + c_3c_1^2 + 3c_2^2c_1 - c_2c_1^3\right) \\
&\vdots
\end{aligned}$$

The reader should already notice how the Riemann Roch for curves and surfaces has part the "geometric" component captured within the 1st and 2nd Todd polynomial.

Again, by substituting $c_i = \binom{n+i}{i}$ defined by:

$$1 + c_1x + \cdots + c_nx^n = (1+x)^{n+1} \bmod x^{n+1}$$

gives

$$T_n(c_1, ..., c_n) = 1$$

Just like before, $Q(x)$ is the only power series with rational coefficients with the property that $Q(x))^{k+1} = 1$ (see [7, p.14], or just repeat the above results with the appropriate modifications).

Recall the Euler characteristic is called the *arithmetic genus* (following Hartshorne's vocabulary). This will motivate how evaluating characteristic series tangent bundles by selecting their Pontryagin/Chern classes shall be called the *genus* with respect to the characteristic series. Depending on the characteristic series, we shall give a name to the genus.

Definition 6.1.5: K-genus

Let V^n be an oriented n-dimensional manifolds and $\{K_j(p_1, ..., p_j)\}$ be an m-sequence. Then:

$$\begin{cases} 0 & \dim_{\mathbb{R}} X \not\equiv 0 \bmod 4 \\ K_n(p_1, ..., p_n)[X] & \dim_{\mathbb{R}} X = 4n \end{cases}$$

is called the *K-genus* of V^n, where $K_n(p_1, ..., p_n)[X]$ represents taking the tangent bundle TM, finding the Pontryagin classes, and evaluating at the fundamental class X.

If we are working with Chern classes, then we would have $n = 2k$.

By what we've discussed earlier, the K-genus is multiplicative.

Proposition 6.1.6: K-genus is Multiplicative

$$K(M \times N) = K(M)K(N)$$

Proof **:**
immediate form properties of characteristic series.

6.1.1 Generating Function

One of the key ideas is to not directly work with the Euler characteristic or Todd class, but a generating function that combines the values. The idea is that we can figure out the coefficients of this generating function when plugin in different values of the dummy variable. To be precise, we shall have the generating function for the Todd class and Euler characteristic, they shall be equal when the dummy variable $y = 1$ as they will both be equal to the index, and when $y = -1$ we shall have that they are equal to their "usual" values, namely the arithmetic and Todd genus.

Let $B = \mathbb{Q}[y]$. Then consider the m-sequence $T_j(y; c_1, ..., c_j)$ with characteristic power series:

$$Q(y; x) = \frac{x(y+1)}{1 - e^{-x(y+1)}} - yx = \frac{x(y+1)}{e^{x(y+1)} - 1} + x$$

Lemma 6.1.7

For every n the coefficient of x^n in $(Q(y; x))^{n+1}$ is equal to $\sum_{i=0}^{n}(-1)^i y^i$, and $Q(y, x)$ is the only power series with coefficients in $\mathbb{Q}[y]$ which has this property.

Proof **:**
Take substitution

$$c_i = \binom{n+1}{i}$$

then:

$$T_n(y; c_1, \ldots, c_n) = 1 - y + y^2 - \cdots + (-1)^n y^n.$$

Then as $Q(\frac{1}{y}; yx) = Q(y; -x)$:

$$y^n T_n(\frac{1}{y}; c_1, \ldots, c_n) = (-1)^n T_n(y; c_1, \ldots, c_n)$$

as we sought to show

The polynomial $T_n(y; c_1, \ldots, c_n)$ can be written in a unique way in the form

$$T_n(y; c_1, \ldots, c_n) = \sum_{p=0}^{n} T_n^p(c_1, \ldots, c_n) y^p.$$

The polynomials $T_n^p(c_1, \ldots, c_n)$ satisfy

$$T_n^p(c_1, \ldots, c_n) = (-1)^n T_n^{n-p}(c_1, \ldots, c_n)$$

Now, consider the factorization:

$$1 + c_1 x + \cdots + c_n x^n = \prod_{i=1}^{n}(1 + \gamma_i x)$$

Then we have:

Proposition 6.1.8: Computing T_y-Polynomial

$$T_n^p(c_1, ..., c_n) = \kappa_n \left[\sum \exp(-\gamma_{j_1} - \cdots - \gamma_{j_r}) \prod_{i=1}^{n} \frac{\gamma_i}{1 - \exp(-\gamma_i)} \right]$$

where the sum is over the $\binom{n}{p}$ pairwise combinations of distinct γ_i, and $\kappa_n[-]$ denotes the sum of all homogeneous polynomials of degree n (mirroring evaluating at the fundamental class in section ref:HERE).

Proof **:**
Indeed;

$$\begin{aligned}
\sum_{p=0}^{n} T_n^p y^p &= \kappa_n \left[\prod_{i=1}^{n} (1 + y\exp(-\gamma_i)) \frac{\gamma_i}{1 - \exp(-\gamma_i)} \right] \\
&= \kappa_n \left[\prod_{i=1}^{n} \frac{(1 + y\exp(-(1+y)\gamma_i))}{1+y} \cdot \frac{(1+y)\gamma_i}{1 - \exp(-(1+y)\gamma_i)} \right] \\
&= \kappa_n \left[\prod_{i=1}^{n} Q(y; \gamma_i) \right] = \sum_{p=0}^{n} T_n^p(c_1, \ldots, c_n) y^p
\end{aligned}$$

as we sought to show

Finally note that $Q(0;x) = \frac{x}{1-e^{-x}}$, $Q(-1;x) = 1 + x$ and $Q(1;x) = \frac{x}{\tanh x}$. Therefore (see 1.5 and Lemma 1.3.1)

$$T_n^0(c_1, \ldots, c_n) = T_n(c_1, \ldots, c_n) \text{ (Todd polynomial)},$$

$$\sum_{p=0}^{n} (-1)^p T_n^p(c_1, \ldots, c_n) = c_n,$$
$$\sum_{p=0}^{n} T_n^p(c_1, \ldots, c_n) = \tilde{L}_n(c_1, \ldots, c_n), \text{ i.e.}$$
$$\sum_{p=0}^{n} T_n^p(c_1, \ldots, c_n) \begin{cases} 0 \text{ if } n \text{ is odd} \\ L_k(p_1, \ldots, p_k) \text{ if } n = 2k. \end{cases}$$

1.9. The Todd polynomials are essentially the Bernoulli polynomials of higher order defined by Nörlund (see N. E. Nörlund, Differenzenrechnung, Berlin, Springer-Verlag, 1924, especially p. 143). The Bernoulli polynomial $B_j^{(n)}(\gamma_1, \ldots, \gamma_n)$ is defined by

$$\prod_{i=1}^{n} \frac{\gamma_i x}{\exp(\gamma_i x) - 1} = \sum_{j=0}^{\infty} \frac{x^j}{j!} B_j^{(n)}(\gamma_1, \ldots, \gamma_n).$$

If the c_i are regarded as the elementary symmetric functions in $\gamma_1, \dots, \gamma_n$ [see 1.8 (14)] then

6.2 Todd Class

With the algebraic established, we can formally define the Todd class:

Definition 6.2.1: Todd Class

Let X be a complex smooth manifold, and ξ a continuous q-vector bundle on X. Let $c_i \in H^{2i}(X, \mathbb{Z})$ be its Chern classes. Then we can define the *total Todd class* of ξ by taking the Todd polynomials and taking:

$$\mathrm{td}(\xi) = \sum_{j=1}^{\infty} T_j(c_1, ..., c_j)$$

where $\{T_j(c_1, ..., c_j)\}$ is given by the characteristic series:

$$Q(x) = \frac{x}{1-e^{-x}} = 1 = \frac{1}{2}x + \sum_{j=1}^{\infty}(-1)^{j-1}\frac{B_j}{(2j)!}x^{2j}$$

where B_j is the Bernoulli numbers.

The reader can verify that the direct product of manifolds $X \times Y$ is multiplicative on the Todd class. This means that if

$$0 \to \xi' \to \xi \to \xi'' \to 0$$

Then:

$$(1 + c_1't + \cdots + c_{q'}'t^{q'})(1 + c_1''t + \cdots + c_{q''}''t^{q''}) = 1 + c_1 t + \cdots + c_q t^q$$

and so:

$$\mathrm{td}(\xi)(t) = \mathrm{td}(\xi')(t) \cdot \mathrm{td}(\xi'')(t)$$

Mirroring the property of the Euler Characteristic.

To write down the m-sequence, recall that the m-sequence of the characteristic power series $Q(z) = \frac{2\sqrt{z}}{\sinh(2\sqrt{z}}$ is has as the first few terms:

$$A_1 = \frac{-2}{3},\ A_2 = \frac{2}{45}(-4p_2 + 7p_1^2)\ A_3 = \frac{-4}{3^2 \cdot 5 \cdot 7}\left(16p_3 - 44p_2p_1 + 31p_1^3\right)$$

Then note that :

$$\frac{x}{1-e^{-x}} = \exp\left(\frac{1}{2}x\right)\frac{\frac{1}{2}x}{\sinh\frac{1}{2}x}$$

By our earlier work:

$$\mathrm{td}(\xi \oplus \xi') = \mathrm{td}(\xi)\mathrm{td}(\xi')$$

In the special case where $q = 1$ and $c_1(\xi) = d \in H^2(X, \mathbb{Z})$, then:

$$\mathrm{td}(\xi) = \frac{d}{1-e^{-d}}$$

Then for a general q-bundle we can apply the splitting principle. In terms of the Todd character, we get define:

$$\sum_{j=0}^{q} c_j x^j = \prod_{i=1}^{q}(1+\gamma_i x)$$

Thus, when dealing with a general complex vector bundle ξ and take its Chern roots γ_i:

$$\mathrm{td}(\xi) = \prod_{i=1}^{q} \frac{\gamma_i}{1-e^{-\gamma_i}}$$

Let us compare this to the usual results we get about the Chern class: it is the natural "dual" of the Todd class in the following way:

Theorem 6.2.2: Relating Todd Class and Chern Character

Let ξ be a continuous $\mathrm{GL}_q(\mathbb{C})$-bundle over X. Then:

$$\sum_{r=0}^{q}(-1)^r \mathrm{ch}\left(\bigwedge^r \xi^*\right) = (\mathrm{td}(\xi))^{-1} c_q(\xi)$$

***Proof* :**
Proof: If $\sum_{j=0}^{q} c_j(\xi)x^j = \prod_{i=1}^{q}(1+\gamma_i x)$ by the properties of Chern classes (though, I did not actually prove this result in my vect. bundle notes. thm 4.4.3 in Hirzebruch, p.64)

$$\mathrm{ch} \bigwedge^r \xi^* = \sum e^{-(\gamma_{i_1}+\cdots+\gamma_{i_r})}$$

where the sum is over all combinations $i_1, \ldots, i_r$ with $1 \le i_1 < \cdots < i_r \le q$. Therefore

$$\sum_{r=0}^{q}(-1)^r \mathrm{ch} \bigwedge^r \xi^* = \prod_{i=1}^{q}(1-e^{-\gamma_i}) = (\gamma_1 \cdots \gamma_q) \prod_{i=1}^{q} \frac{1-e^{-\gamma_i}}{\gamma_i} = (\mathrm{td}(\xi))^{-1} c_q(\xi).$$

6.2.3 Relation to Serge Class There seems to be a relation to Serge class I wouldn't mind pointing out

Recall that if we are talking about the Chern class of a complex manifold M, we are talking about the Chern class of its tangent bundle, and the K-genus is defined for such structures:

Definition 6.2.4: Todd Genus

Let M be an n-dimensional almost complex manifold. The *Todd genus* (or *T-genus*) is the number:

$$T(M_n) = T_n(c_1, ..., c_n)[M_n] = \kappa_n[\mathrm{td}(TM)] = \int_M \mathrm{td}(TM)$$

where $[M_n]$ represents the tangent bundle of M_n.

The *generalized Todd Genus* is given by $\{T_y(y; c_1, ..., c_n\}$ and:

$$T_y(M_n) = \sum_{p=0}^{n} T^p(M_n) y^p$$

is the *generalized Todd Genus.*

The generalized Todd genus is a natural generaliztion of the todd genus as:

$$T_0(M_n) = T^0(M_n) = T(M_n)$$

To important terms that we must evaluate is at the $y = -1$ and $y = 1$. For $y = -1$:

$$\begin{aligned}
T_{-1}(M_n) &= \sum_{p=0}^{n} (-1)^p T^p(M_n) \\
&= \prod_i Q(-1; \gamma_i)[M_n] \\
&= \prod_i Q(1 + \gamma_i x)[M_n] \\
&= c[M_n] \\
&\stackrel{!}{=} c_n[M_n] \\
&= \chi(M_n)
\end{aligned}$$

where $\stackrel{!}{=}$ comes from ignoring the lower order terms because perfect pairing. Evaluating at 1 we get:

$$T_1(M_n) = \sum_{p=0}^{n} T^p(M_n) = \tau(M_n)$$

where τ is the index of M_n, with the equality holding since

$$Q(1; x) = \frac{x(2e^{2x} - (e^{2x} - 1))}{e^{2x} - 1} = \frac{x(e^{2x} + 1)}{e^{2x} - 1} = x \coth(x) = \frac{x}{\tanh(x)}$$

This shall be important for the eventual comparison of with the Euler characteristic. By lemma 6.1.7, the T-genus is the only genus associated to an m-sequence with rational coefficients which takes the value 1 on every complex projective space $\mathbb{C}\mathrm{P}^n$, and the T_y-genus is the only genus associated to an m-sequence with coefficients in $\mathbb{Q}[y]$ which takes the value $1 - y + y^2 - \cdots + (-1)^n y^n$ on $\mathbb{C}\mathrm{P}^n$.

Proposition 6.2.5: Properties of Todd Genus

Let B be a commutative ring where $Q \subseteq B$. Then

1. $\sum_{l=0}^{n} T_n^{(l)}(c_1, \ldots, c_n) = c_n$, for all n.
2. $T_n^{(0)}(c_1, \ldots, c_n) = \mathrm{td}(c_1, \ldots, c_n)(= T_n(c_1, \ldots, c_n))$.
3. $$T_n^l(c_1, ..., c_n) = (-1)^n T_n^{n-l}(c_1, ..., c_n)$$
4. $$\sum_{l=0}^{n} T_n^{(l)}(c_1, \ldots, c_n) = \begin{cases} 0 & \text{if } n \text{ is odd} \\ L_{\frac{n}{2}}(p_1, \ldots, p_{\frac{n}{2}}) & \text{if } n \text{ is even.} \end{cases}$$

The *total Todd class* of a vector bundle, ξ, is

$$\mathrm{td}(\xi)(t) = \sum_{j=0}^{\infty} \mathrm{td}_j(c_1, \ldots, c_j)t^j = 1 + \frac{1}{2}c_1(\xi) + \frac{1}{12}(c_1^2(\xi) + c_2(\xi))t^2 + \frac{1}{24}(c_1(\xi)c_2(\xi))t^3 + \cdots$$

Proof **:**
exercise

6.3 T-characteristic – combining Algebraic and Geometric information

So far, we saw that the Todd character captures some information about the underlying space X. We shall want to capture some information about the bundle as well; this is given by the Chern character. Combining these two together, we get:

Definition 6.3.1: T-characteristic

Let X be an admissible complex manifold, ξ a $U(q)$-vector bundle. Then

$$T(X, \xi)(t) = \kappa[\mathrm{ch}(\xi)\mathrm{td}(TM)] = \int_X \mathrm{ch} \cdot \mathrm{td}$$

Is called the *T-characteristic* of ξ over X

Note that if $\xi = \mathbb{1}$, then $T(X, \mathbb{1}) = \kappa[\mathrm{ch}(\mathbb{1}\mathrm{td}(TM)] = \kappa[\mathrm{td}(TM)]$ showing that we reduce to only holding information about X. As $\mathbb{C}^*$-bundles over M_n are classified by elements $d \in H^2(M_n, \mathbb{Z})$, we shall sometimes write $T(M_n, d)$ for $T(M_n, \xi)$ where ξ is associated to d. A thing with virtual, eq. 4 (WILL BE MOVED TO VIRTUAL SECTION .

If ξ is a complex vector bundle of dimension q, ξ' of dimension q', both over M_n, then (by the virtual equation of eq4) we get:

$$T(M_n, \xi \oplus \xi') = T(M_n, \xi) + T(M_n, \xi')$$

Then by the properties of the characters we get:

$$T(M \times N, \pi_M^*(\xi) \otimes \pi_N^*(\eta)) = T(M, \xi)T(N, \eta)$$

Let us extend this to T_y-characteristics and show that it is additive and multiplicative in the same way. First, by proposition 6.1.8:

$$T\left(M_n, \bigwedge^p (T^*M)\right) = T^p(M_n)$$

Then $\wedge^p(T^*M) \otimes \xi$, so $T(M_n, \lambda^p(T^*M)) = T^p(M_n)$. By taking:

$$T_y(M_n, \xi) = \sum_{p=0}^{n} T^p(M_n, \xi)y^p$$

we with (virtual eq.4) we get:

$$T(M_n, \xi) = \kappa_n \left[\mathrm{ch}(\xi)\mathrm{ch}\left(\bigwedge^p (T^*M)\right) \mathrm{td}(TM)\right]$$

Then by applying the same proof techniques for proposition proposition 6.1.8 we get:

$$T_y(M_n, \xi) = \kappa_n \left[\left(\sum_{i=1}^{q} e^{(1+y)d_i}\right)\left(\sum_{j=0}^{n} T_j(y; c_1, \ldots, c_j)\right)\right]$$

Notice that:

$$T_{-1}(M_n, \xi) = q \cdot \chi(M_n)$$

recovering the Euler-Poincaré characteristic of M_n. Similarly, generalizing proposition 6.2.5 we get:

$$T^p(M_n, \xi) = (-10^n T^{n-p}(M_n, \xi^*)$$

and if $p = 0$:

$$T(M_n, \xi) = (-1)^n T\left(M_n, \bigwedge^n (T^*M) \otimes \xi^*\right)$$

Then, by doing the same techniques to generalize the Total Chern class to the Total y-Chern class, we can se that:

$$T_y(M_n, \xi \oplus \xi') = T_y(M_N, \xi) + T_y(M_n, \xi')$$

and

$$T_y(M \times N, \pi_M^*(\xi) \otimes \pi_N^*(\eta)) = T_y(M, \xi)T_y(N, \eta)$$

6.4 Split Manifolds and Splitting Method

As far as I can tell, this is just explaining the splitting method and so I don't need to go over this because I already do in [5]

I will look at Gallier notes later to see if there is anything I need to fill in here

Move 12.3 here more virtual results

6.4.1 Virtual Todd Genus

Recall the if E a vector bundle over a manifold M, there is a flag manifold $F(M)$ such that the pullback along $f : F(M) \to M$ makes E split into linear bundles. The same applies even if $M = V$ is a variety with singularities; we can breakdown the pullback of E into a filtration:

$$f^*(E) = E_r \supseteq E_{r-1} \supseteq \cdots \supseteq E_1 \supseteq E_0 = 0$$

This shall afford us the ability to calculate the Todd character, and hence the Todd genus, with greater ease. We shall not cover all the details, they can be found in [9] or [7].

Let us start by relating the Chern character to a sub-manifold Let $N_{n-k} \subseteq M_n$ be a submanifold of codimension k and let $j : V_{n-k} \to M_n$ be an embedding. If M_n is almost complex, then we have the pullback:

$$j^*(TM_n) = TN_{n-k} \oplus N_{M/N} \tag{6.1}$$

where $N_{M/N}$ is the normal bundle of N with respect to M; let us label it ν. if $k = 1$, then we simply get:

$$c(\nu) = 1 + j^*(v)$$

where $v \in H^2(M_n; \mathbb{Z})$ is the cohomology class determined by the fundamental calls of V_{n-1}. By equation (6.1):

$$\begin{aligned} &1 + c_1(V_{n-1}) + c_2(V_{n-1}) + \cdots \\ &= j^* \left((1 + c_1(M_n) + c_2(M_n) + \cdots)(1 + v)^{-1}\right) \end{aligned}$$

Then we can define the T_y-genus of V_{n-1}. To that end, define the characteristic power series:

$$Q(y; x) = \frac{x}{R(y; x)}$$

where $R(y; x)$ is a shorthand for $\frac{e^{x(y+1)}-1}{e^{x(y+1)}+y}$. Then equation (6.1) implies:

$$\sum_{i=0}^{\infty} T_i(y; c-1(V_{n-1}), ..., c_i(V_{n-1})) = j^* \left(\frac{R(y; v)}{v} \sum_{i=0}^{\infty} T_i(y; c_1(M_n), ... \right)$$

and so the T_y-genus is:

$$T_y(V_{n-1}) = \kappa_n \left[R(y; v) \sum_{j=0}^{\infty} T_j(y; c_1(M_n), ..., c_j(M_n)) \right] = \kappa_n[(1 - e^{-v})\mathrm{td}(TM)]$$

where the 2nd equality is an immediate application of equation (6.1). Now, by the splitting principal, we can filter TM, or any vector bundle E, where each intermediate step is linear. This motivates the following definition:

Definition 6.4.1: Virtual Todd Genus

Let $v_1, ..., v_r \in h^2(M_n; \mathbb{Z})$. Then:

$$T_y(v_1, ..., v_r)_M = \kappa_n \left[R(y; v_1) \cdots R(y; v_r) \sum_{j=0}^{\infty} T_j(y; c_1(M_n), ...) \right]$$

where the subscript denotes what is the original manifold, and will be omitted if it is clear from context.

By proposition 6.1.8, the virtual Todd genus is a polynomial of degree $n - r$ in y with rational coefficients. if $r > n$, it is 0, and if and only if $r = n$:

$$T_y(v_1, ..., v_n)_M = v_1 v_2 \cdots v_n [M_n]$$

Hirzebruch now takes the time to relate the fundamental classes v_i in M to the pushforward via j^*; detail in 11.2/11.3

Let us now develop this to the define the virtual T-characteristic

this is 12.3 of Hirzebruch. 12.3 is the last of section 12

6.4.2 Application to Splitting Principle

The important thing is when the Todd genus comes back into the discussion. I will put here the last section, 13.6:

13.6. Let X be a compact almost complex split manifold of complex dimension n. We use the notations of 13.5 a) and derive a formula which will imply that the TODD genus $T(X)$ can be expressed in terms of virtual indices:

$$(1+y)^n T(X) = \sum_{l=0}^{n} y^l \sum_{1 \le i_1 < \cdots < i_l \le n} T_y(a_{i_1}, \ldots, a_{i_l})_X \tag{13}$$

For the proof we recall the definition of the virtual T_y-genus in 11.2 (7), apply (12), and obtain for the right hand side of (13)

$$\begin{aligned}
\kappa_n \left[\prod_{i=1}^{n} (1 + yR(y; a_i)) \prod_{i=1}^{n} Q(y; a_i) \right] &= \kappa_n \left[\prod_{i=1}^{n} (Q(y; a_i) + a_i y) \right] \\
&= \kappa_n \left[\prod_{i=1}^{n} \frac{(1+y)a_i}{1 - \exp(-(1+y)a_i)} \right] \\
&= (1+y)^n \kappa_n \left[\prod_{i=1}^{n} \frac{a_i}{1 - \exp(-a_i)} \right] = (1+y)^n T(X)
\end{aligned}$$

For $y = 1$ the virtual T_y-genus becomes the virtual index:

$$2^n T(X) = \sum_{l=0}^{n} \sum_{1 \le i_1 < \cdots < i_l \le n} \tau(a_{i_1}, \ldots, a_{i_l})_X \tag{13*}$$

Since the virtual index is an integer (Theorem 9.3.1) a corollary is

Theorem 13.6.1. *The TODD genus of a compact almost complex split manifold multiplied by 2^n is an integer.*

Formula (13) can be generalised to apply to the virtual T-genus. If $b_1, \ldots, b_r \in H^2(X, Z)$, $(r \le n)$, then

$$(1+y)^{n-r} T(b_1, b_2, \ldots, b_r)_X = x_n \left[\prod_{i=1}^{r} R(y; b_i)(1 + yR(y; b_i))^{-1} \prod_{j=1}^{n} (1 + yR(y; a_j)) \prod_{k=1}^{n} Q(y; a_k) \right] \tag{14}$$

This formula, and the definition of the virtual T_y-genus, imply that $(1+y)^{n-r}T(b_1, \ldots, b_r)_X$ can be expressed as a sum of terms each of which is a virtual T_y-genus multiplied by a polynomial in y with integral coefficients. If $y = 1$ then $2^{n-r}T(b_1, \ldots, b_r)_X$ is expressed as a sum of

Theorem 6.4.2: Splitting and Virtual Todd Genus

Let X be a compact almost complex split manifold, and let $b_1, \ldots, b_r$ be elements of $H^2(X, \mathbb{Z})$. The virtual TODD genus $T(b_1, \ldots, b_r)_X$ multiplied by 2^{n-r} is an integer.

6.4.3 Todd Genus Invariant Under Splitting

Let us now show that $T(X) = T(X_\Delta)$, that is the Todd genus is invariant under splitting. Essentially, $T(F(n)) = 1$ and $T(X_\Delta) = T(X)T(F(n)) = T(X)$.

7

Hirzebruch Riemann Roch

here

7.1 Classical Riemann Roch Proofs

(This may not be important for us; it will give the fact that $\dim_{\mathbb{C}} H^i = 0$ for certain i.)
X is admissible if:

1. X is σ-compact (locally compact and a countable union of compacts)
2. X has finite combinatorial dimension X

As (1) implies paracompact, our work on fine and soft sheaves goes through.

Theorem 7.1.1: RR For Riemann Surface with Line Bundles

Let X be a compact Riemann surface and $\mathcal{L}$ a complex analytic line bundle on X. Then

1. There exists a Cartier divisor D such that $\mathcal{L} \cong \mathcal{O}_X(D)$ so that $\deg(\mathcal{L}) := \deg(D)$
2. The Riemann-Roch equation holds:
$$\dim_{\mathbb{C}} H^0(X, \mathcal{L}) - \dim_{\mathbb{C}} H^0(X, \omega_X \otimes \mathcal{L}^{\vee}) = \deg(\mathcal{L}) + 1 - g$$
where (by ref:HERE, or [1, section 5.3]) $g = \dim H^0(X, \omega_X) = \dim H^1(X, \mathcal{O}_X)$ is the genus of X.

***Proof* :**
The first point is an application of much proved before: By ref:HERE X is an algebraic variety, in particular a curve, and by [1, section 6.1] there is an embedding $X \hookrightarrow \mathbb{P}^N_{\mathbb{C}}$ for some N. Then by ref:HERE (GAGA, which I'll prove) we have that $\mathcal{L}$ is isomorphically pushed forward through these maps hand is an algebraic line bundle. As X is a variety, by [1, section 4.3] there exists a Cartier divisor D such that $\mathcal{L} = \mathcal{O}_X(D)$. Let us show $\deg(\mathcal{L}) = \deg(D)$ is well-defined, i.e. independent of linear equivalence. Let $f \in \mathcal{M}(X)$. Then as shown in [17, section 6.3] there exists a branch covering $f : X \to \mathbb{P}^1_{\mathbb{C}}$ where:

$$\#(f^{-1}(0) = \#(f^{-1}(\infty)) = \text{deg. of map}$$

Hence:

$$\deg(f)\#(f^{-1}(0) - \#(f^{-1}(\infty)) = 0$$

Thus, $D \sim E$ implies $\deg(D) = \deg(E)$, hence $\deg(\mathcal{L})$ is well-defined

For the 2nd point, if $D = 0$, then $\mathcal{O}_X(0) = \mathcal{O}_X$, and $H^0(X, \mathcal{O}_X) = \mathbb{C}$ so $l(0) = 1$ and $i(0) = g$ and

$$l(0) - i(0) = 1 - g = \deg(0) + 1 - g$$

by [17, section 6.3], hence assume $D \neq 0$ (though the beginning of this discussion doesn't need this assumption). First use Serre Duality to get:

$$H^0(X, \omega_X \otimes \mathcal{L}^\vee) \cong H^1(X, \mathcal{L})^\vee$$

Reducing the right hand side to the Euler characteristic $\chi(X, \mathcal{O}_X(D))$. To get the right hand side, we take advantage of additivity of χ. For any point $P \in X$, we get an exact sequence:

$$0 \to \mathcal{O}_X(-P) \to \mathcal{O}_X \to \kappa_P \to 0$$

when κ_P is the skyscraper sheaf:

$$(\kappa_P)_x = \begin{cases} (0) & x \neq P \\ \mathbb{C} & x = P \end{cases}$$

Note that as this sheaf is flasque, H^i disappears for $i > 0$ so $\chi(X, \kappa_P) = 1$.

Go over $\chi(X, \kappa_D)$

Tensoring with $\mathcal{O}_X(D)$, and applying our theory from [1, section 4.3] we get:

$$0 \to \mathcal{O}_X(D - P) \to \mathcal{O}_X(D) \to \kappa_P \otimes \mathcal{O}_X(D) \to 0$$

Thus, by additive of χ:

$$\chi(X, \mathcal{O}_X(D) = \chi(X, \kappa_P \otimes \mathcal{O}_X(D)) + \chi(X, \mathcal{O}_X(D - P))$$

Now, if $D > 0$, we can pick a point P appearing in D. Then $\deg(D-P) = \deg(D) - 1$. We can thus do induction on $\deg(D)$. If $\deg(D) = 0$, then $\mathcal{O}_X(D)$ has holomorphic functions, which as X is compact is only constant functions, and by the Brill-Noether Reciprocity ([17]) we have $i(D) = g$, giving the base case. Now suppose the result holds for $\deg(D) = n - 1$ and consider $\deg(D) = n$. Then $\deg(D - P) = n - 1$ from which we get by the inductive hypothesis

$$\deg(D - P) + 1 - g = \chi(X, \mathcal{O}_X(D - P))$$

words here; I need χ to split with D and P

$$\deg(D) + (-1+1) + 1 - g = \chi(X, \mathcal{O}_X(D))$$

giving the desired result.

For arbitrary D, we can split $D = D^+ - D^-$ where $D^+, D^- \geq 0$. Then we get a short exact sequence:

$$0 \to \mathcal{O}_X \to \mathcal{O}_X(D^-) \to \kappa_{D^-} \to 0$$

Then by additivity;

$$\chi(X, \mathcal{O}_X(D^-)) = \chi(X, \mathcal{O}_X) + \chi(X, \kappa_{D^-}) = 1 - g + \deg(D^-)$$

Now, tensor with $\mathcal{O}_X(D)$ to get the short exact sequence:

$$0 \to \mathcal{O}_X(D) \to \mathcal{O}_X(D + D^-) \to \kappa_{D^-} \to 0$$

where the last term is the same (as can be verified point-wise). Applying additivity of χ again we get:

$$\chi(X, \mathcal{O}_X(D + D^-)) = \chi(X, \mathcal{O}_X(D)) + \chi(X, \kappa_{D^-}) = \chi(X, \mathcal{O}_X(D)) + \deg(D^-)$$

As $D + D^- = D^+$, re-arranging we get:

$$\chi(X, \mathcal{O}_X(D)) = -\deg(D^-) + \chi(X, \mathcal{O}_X(D^+))$$

Applying our earlier result we get:

$$\chi(X, \mathcal{O}_X(D)) = -\deg(D^-) + \deg(D^+) + 1 - g \deg(D) + 1 - g$$

as we sought to show.

As $\mathcal{L}$ is a line bundle, we can take its value of the first Chern class $c_1(\mathcal{L}) \in H^2(X;\mathbb{Z})$. By letting $[X] \in H - 2(X;\mathbb{Z}) \cong \mathbb{Z}$ be the fundamental class, then:

$$\deg(\mathcal{L}) = c(\mathcal{L})[X] \in \mathbb{Z}$$

Then theorem 7.1.1 becomes:

$$\begin{aligned}
\chi(X, \mathcal{L}) &= c_1(\mathcal{L})[X] + 1 - g \\
&= \left[c_1(\mathcal{L}) + \frac{1}{2}(2 - 2g)\right][X] \\
&= \left[c_1(\mathcal{L}) + \frac{1}{2}c_1(T_X^{1,0})\right][X]
\end{aligned}$$

By Bass-Serre Cancellation (see [5, section 5.1]) or by some result from the theory of characteristic (see [5, section 3.2]), higher-dimensional vector bundles are trivial; in particular as $\dim_{\mathbb{C}} X = 1$, any vector bundle of dimension $n > 1$ will have “trivial” information cohomologically. Let us show this explicitly to further motivate HRR. We first require a special case of

This following theorem may also be in my char.class notes we also technically don't need it as we can prove the result for vector bundles via HRR with a lot more theory, but I think this is good to confirm that HRR is correct

Theorem 7.1.2: Atiyah-Serre On Vector Bundles

Let X be either a compact complex manifold or an algebraic variety; in either case of dimension $d = \dim_{\mathbb{C}} X$. Let E be a rank $r > d$ vector bundle on X that is smooth or algebraic whether X is a manifold of variety respectively [a]. Then there is a trivial bundle $\mathbb{I}^{r-d}$ of rank $r - d$ that fits into the short exact sequence:

$$0 \to \mathbb{I}^{r-d} \to E \to E'' \to 0$$

and $\mathrm{rk}(E'') = d$.

[a] If algebraic, assume $\Gamma_{\mathrm{alg}}(X, \mathcal{O}_X(E)) \to E_x$ is surjective for all x;i.e. it is generated by its global sections

***Proof* :**
If E is smooth, then by partitions of unity it is always generated by its global sections. As algebraic vector bundles need not be fine, we must assume they are (for example $\mathcal{O}_{\mathbb{P}^1}(-1)$ does not satisfy this condition). Without loss of generality, we shall work over $\overline{k} = \mathbb{C}$.

Then, for each $x \in E$, $\Gamma(X, \mathcal{O}_X(E)) \twoheadrightarrow E_x$ surjective onto the fiber E_x. Then we can choose a finite dimensional subspace $W(x) \subseteq \Gamma(X, \mathcal{O}_X(E))$ as $W(x) \to E_x$ surjective. Then either $\mathbb{C}$- or Z-near x, $\Gamma(X, \mathcal{O}_X(E))$ surjects onto the fibers in the neighbourhood (namely, the determinant is smooth/regular, and hence we can follow a similar argument in both cases). As X is [quasi]-compact, there exists a finite open cover with this property.

Label these choices $W(i)$. Taking all their generators, we have a finite dimensional subspace $W \subseteq \Gamma(X, \mathcal{O}_X(E))$ such that $W \twoheadrightarrow E_x$ surjects for all $x \in X$ (namely $\sigma \mapsto \sigma(x)$ surjects). Let:

$$\ker(x) = \ker(W \to E_x)$$

Then as $\dim E_x = r$,

$$\dim \ker(x) = \dim W - r$$

As the dimension is constant, $\dim \ker(x)$ is independent of X, thus we can take their union of X. To be precise, let $\mathbb{P}(\ker(x)) \subseteq \mathbb{P}(W)$ be the projectivizations of $\ker(x) \subseteq W$ (i.e. collection of all lines). Then consider

$$Z = \cup_{x \in X} \mathbb{P}(\ker(x))$$

and takes its Z-closure if necessary. Then:

$$\dim Z = \dim X + \dim W - r - 1 = \dim W + d - r - 1$$

where the -1 comes from the fact that $\dim(\mathbb{P}(V)) = \dim V - 1$. Hence, we have:

$$\mathrm{codim}(Z \hookrightarrow \mathbb{P}(W)) = r - d$$

Then it is simple algebraic/affine geometry to see there exists a projective (linear) subspace $T \subseteq \mathbb{P}$ with $\dim T = r - d - 1$ such that

$$T \cap Z = \emptyset$$

Let $T = \mathbb{P}(S)$ for some subspace $S \subseteq W$ (so $\dim S - r - d$). Then by construction:

$$X \times S = X \times \mathbb{C}^{r-d} = \mathbb{I}^{r-d}$$

From this, we can define a map $\mathbb{I}^{r-d} \hookrightarrow E$ sending $(x, s) \in s(x) \in E$. As $T \cap Z = \emptyset$, $s(x)$ is never zero, and hence $\operatorname{im}(\mathbb{I}^{r-d} \hookrightarrow E)$ has full rank at each point. Letting $E'' = E/\operatorname{im}(\mathbb{I}^{r-d} \hookrightarrow E)$, we get a vector bundle of dimension d and a short exact sequence:

$$0 \to \mathbb{I}^{r-d} \to E \to E'' \to 0$$

as we sought to show.

Now, by section 3 we have that $c_i(\mathbb{I}^r) = 0$ for all i, so by additivity of characteristic classes we have that $c_1(E) = c_1(E'')$. As $c_1(E) = c_1\left(\bigwedge^r E\right)$, we get:

$$c_1(E) = c_1\left(\bigwedge^{\mathrm{rk}(E'')} E''\right)$$

Using this, we can prove the following "simple" generalization:

Theorem 7.1.3: RR For Riemann Surface With Vector Bundles

Let X be a compact Riemann surface and E a complex vector bundle over X of rank r. Then:

$$\chi(X, E) = c_1(E) + r(1 - g)$$

whee $c_1(E)$ is the first Chern class of E.

Proof **:**
Start with embedding X in $\mathbb{P}^N$ via an algebraic equation and pushforward E so that it is an algebraic vector bundle. Then E need not be generated by global holomorphic functions, however by twistign it enough

$$E(n) := E \otimes \mathcal{O}_X(n) \cong E \otimes \mathcal{O}_X^{\otimes n}$$

we shall have a very ample sheaf, and hence $E(n)$ shall have enough global section (see [1, section 4.4] for the details). From this, we can apply theorem 7.1.2:

$$0 \to \mathbb{I}^{r-1} \to E(n) \to E'' \to 0$$

where $\mathrm{rk}(E'') = 1$. Then twisting it by $\mathcal{O}_X(-n)$ we get:

$$0 \to \coprod_{r-1} \mathcal{O}_X(-n) \to E \to E''(-n) \to 0 \tag{7.1}$$

Then:

$$\chi(X, \mathcal{O}_X(E)) = \chi(X, E''(-n)) + \chi\left(\coprod_{r-1} \mathcal{O}_X(-n)\right)$$

As $\mathrm{rk}(E''(-n)) = 1$, by ordinary Riemann-Roch we have:

$$\chi(X, E''(-n)) = c_1(E''(-n)) + 1 - g$$

giving us:

$$\chi(X, \mathcal{O}_X(E)) = c_1(E''(-n)) + 1 - g + \chi\left(\coprod_{r-1} \mathcal{O}_X(-n)\right)$$

From this, we shall proceed with inducting on the rank r. The base case $r = 1$ was theorem 7.1.1. Assume Riemann Roch holds for $r-1$, so that by the inductive hypothesis:

$$\chi\left(\coprod_{r-1} \mathcal{O}_X(-n)\right) = c_1\left(\coprod_{r-1} \mathcal{O}_X(-n)\right) + (r-1)(1-g)$$

Then:

$$\chi(X, \mathcal{O}_X(E)) = c_1(E''(-n)) + 1 - g + c_1\left(\coprod_{r-1} \mathcal{O}_X(-n)\right) + (r-1)(1-g)$$

Then we can combine the Chern classes by additivity on equation (7.1), and so;

$$\chi(X, \mathcal{O}_X(E)) = c_1(E) + r(1-g)$$

as we sought to show.

We shall show that:

$$\chi(X, \mathcal{O}_X(E)) = c_1(E) + \frac{\mathrm{rk}(E)}{2} c_1(T_X^{1,0})$$

7.2 Hirzebruch Riemann Roch Theorem

Let M be an algebraic manifold, $T(M)$ be the Todd genus, and $\chi(M)$ be the arithmetic genus. We shall show that $T(V) = \chi(V)$. This then RR for complex line bundles over M.

Theorem 7.2.1: Splitting of Algebraic Manifold

Let M be an algebraic manifold which is also a complex split manifold. Then:

$$\chi(M) = T(M)$$

***Proof* :**
Let $m = \dim M$, and $A_1, ..., A_m$ the complex diagonal line bundles defined over M. Then by section 13.6 eq.(13):

$$(1+y)^m T(M) = \sum_{l=0}^{m} y^l \sum_{1 \le i_1 < \cdots < i_l \le n} T_y(a_{i_1}, ... a_{i_l})_M$$

where the T_y are polynomials, and by theorem 17.4.1 (in virtual char:

$$(1+y)^m \chi(V) = \sum_{l=0}^{m} y^l \sum_{i_1 < \cdots < i_l} \chi_V(A_{i_1}, \ldots, A_{i_l})_V$$

As M is algebraic, theorem 19.2.1 (vert char for alg man) χ_y is a polynomial. Then we see by the equation that the two agree if for some $y \neq -1$ (or else $1 + y = 0$) the two match. By choosing $y = 1$, theorem 19.5.1, literally last result of the section say they agree at $y = 1$, completing the proof.

Now, as we know we can pullback any vector bundle E over M to via the map $F(M) \to M$ where $F(M)$ is the flag manifold, by theorem 14.3.1 (saying the T-char equals on pullback we have $T(M) = T(F(M))$ (Hirzebruch denotes $F(M)$ as M^{Δ}). We just now need the corresponding result for the arithmetic genus:

Theorem 7.2.2: Euler-characteristic Invariant Under Splitting

Let ξ be a complex $\mathrm{GL}_q(\mathbb{C})$-bundle over an algebraic manifold M, $M' = F(M)$ the flag manifold. Let φ be the map and $\varphi^*(\xi)$ be the pullback. Then

$$\chi(M) = \chi(M')$$

Now I'm a bit confused, why no ξ, $\varphi^*(\xi)$ with the Eul.Char???

Proof **:**
[7, p.149], quote:

1. theorem 18.3.1*
2. theorem 15.9.1

Theorem 7.2.3: Todd and Arithmetic Genus Equivalence

Let M be an algebraic manifold. Then:

$$\chi(M) = T(M)$$

Proof **:**
string the above together!

The above can be thought as proving that $\chi_0(X) = T_0(M)$. Then theorem 19.5 implies:

Theorem 7.2.4: Extension To Line Splitting

Let M be an algebraic manifold, $F_1, ..., F_r$ complex line bundles over M. Then:

$$\chi(F_1, ..., F_r)_M = T(F_1, ..., F_r)_V$$

Proof **:**
This is quoting theorem 19.5

Note for $r = 1$, $F_1 = F$, $\chi(F)_M = T(F)_V$. Then combining theorem 17.2 and theorem 12.1(4):

$$\chi(F) = \chi(M) - \chi(M, F^{-1})$$

and:

$$T(F) = T(M) - T(M, F^{-1})$$

replacing F by F^{-1} theorem 20.2.2, theorem 20.3.1 implies:

$$\chi(M, F) = T(M, F)$$

But this *is* HRR for line bundles!! I will box this to make it obvious.

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